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Quality of life in breast cancer patients with neoadjuvant chemotherapy: a systematic review

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Quality of life in breast cancer patients with neoadjuvant chemotherapy: a systematic review

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Abstract

Objectives Neoadjuvant chemotherapy (NAC) shrinks tumors and improves survival in breast cancer (BC) patients. But it can also have an impact on patients' quality of life (QOL). Therefore, it is important to assess the QOL of BC patients during and after NAC. The aims of this systematic evaluation study were (1) to assess the impact of NAC on BC patients' QOL; (2) to compare the different regimens of NAC on BC patients' QOL; (3) to compare NAC versus adjuvant chemotherapy on BC patients' QOL; (4) to identify predictors of QOL on BC patients receiving NAC.

Methods We performed a systematic review of literature using the PRISMA guidelines. Cinahl, Embase, Pubmed, Scopus, Web of Science, Cochrane library, and PsycINFO databases were searched.

Results The initial search resulted in 2994 studies; 12 of these studies fulfilled our inclusion criteria. There was no significant difference in the QOL of BC before and after NAC, but patients experienced adverse reactions and depression during chemotherapy. Different regimens of NAC have different effects on patients' QOL. Patients with NAC had more severe physical discomfort than those with adjuvant chemotherapy. However, BC patients' QOL can be improved by intervening on social or family support and these predictors including chronotype, QOL before NAC and depression.

Conclusion More original research is needed in the future to understand the profile and predictors of QOL in BC patients on NAC, which will help clinicians and patients make decisions and deal with NAC-related issues.

Article Summary

Strengths and limitations of this study

This systematic review was designed and reported as PRISMA guidelines.

This review studied the impact and predictors of neoadjuvant chemotherapy on the quality of life of breast cancer patients.

The electronic search will only include studies published in English that could limit the inclusion of studies.

Keywords Breast cancer; Neoadjuvant chemotherapy; Quality of life; Systematic review

Introduction

Breast cancer (BC) is the most common disease in women worldwide, with the prevalence now surpassing lung cancer as the leading global cancer incidence in women in 2020, with 11.7% of all new cases (2.3 million); and it is the fifth leading cause of cancer deaths worldwide, accounting for 6.9%[1]. Treatment of BC usually consists of two parts: local surgical treatment and adjuvant systemic treatment; systemic treatment can precede surgical treatment (neoadjuvant) or follow it (adjuvant)[2]. Neoadjuvant chemotherapy (NAC) is used in patients with locally advanced BC or inoperable BC to improve the likelihood of breast-conserving surgery by reducing the size of the tumor in the past. Current studies have shown that NAC can also be used in the treatment of patients with early-stage operable BC to improve cosmetic outcomes and reduce postoperative complications such as lymphedema and prolong survival[3–5]. NAC is associated with improved survival compared with adjuvant chemotherapy in patients with triple-negative BC only after complete pathologic response[6], and the same effect has been observed in other cancers like bladder cancer and colon cancer[7,8]. NAC is widely used, while more and more scholars are studying the survival outcomes of patients after NAC, such as pathologic complete response, disease-free survival and overall survival[4,9].

However, the goals of cancer treatment should include not only survival outcomes but also quality of life (QOL), which is an important outcome measure in clinical investigations and survival studies in BC[10–12]. Now the treatment regimens of BC are numerous and effective, but the side effects associated with different treatment modalities and the impact on QOL are different. So optimal stratification and ranking of treatment strategies should focus on individual risk profiles as well as QOL[13]. Meanwhile, QOL is a major factor in treatment decisions for advanced or metastatic disease[14]. It is worthwhile to consider the choice of an effective, least toxic anticancer treatment before implementing definitive therapy. Therefore, studying the QOL of NAC patients and comparing it to other treatments can help patients and clinicians make decisions.

The aims of this systematic evaluation study were (1) to assess the impact of NAC on BC patients' QOL; (2) to compare the different regimens of NAC on BC patients' QOL; (3) to compare NAC versus adjuvant chemotherapy on BC patients' QOL; (4) to identify predictors of QOL on BC patients receiving NAC.

Methods

This systematic review was designed and reported as perPreferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[15].(Figure 1)

Search strategy

A systematic search of the following databases was performed: CINAHL, Embase, Pubmed, Scopus, Web of Science, Cochrane library, and PsycINFO. The final search was in December 2021. Three categories of terms were searched: (1)breast cancer, (2) neoadjuvant chemotherapy, (3) quality of life. In PubMed and Cochrane library, Medical Subject Headings (MeSH) were used (breast neoplasms, neoadjuvant therapy, quality of life, sexuality, anxiety, depression, social support). In Embase, Emtree terms were exploded (breast tumor, neoadjuvant chemotherapy, quality

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of life, sexuality, anxiety, depression, social support). In Scopus, Web of Science, CINAHL, and PsycINFO, only keywords were used.(Supplementary material)

Data analysis

The inclusion criteria were as follows: (1) adult patients (18 years older); (2) BC patients receiving NAC; (3) outcome measures related to QOL, including QOL, psychological impact, support from various sources and so on; (4) published in the English language. The exclusion criteria were as follows: (1) data could not be extracted; (2) duplicates or overlapping participants; (3) not original research, such as commentaries, editorials, or reviews;(4) full text not available;(5) qualitative study.Two independent researchers (YQ Z and XQ Z) screened for potentially relevant studies by reviewing abstracts. The full texts were then screened further according to the inclusion and exclusion criteria. Any uncertainty was solved by a third researcher(YL S). Two researchers (YQ Z and XQ Z) independently extracted and recorded the following information from the enrolled studies: author, published year, country, study design, sample size, age, regimens of NAC, timing of assessment, instrument, outcomes.The risk of bias in individual studies was evaluated by 2 independent researchers (YQ Z and LC C) in accordance with Cochrane collaboration's tool for assessing risk bias[16], or Newcastle Ottawa Score(NOS)[17] or Joanna Briggs Institute Critical Appraisal tool[18]. Any disagreement was resolved by a third researcher(YL S).

Patient and public involvement

Patients or the public were not involved in the conceptualisation or carrying out of this research.

Results

Flow of included studies

An initial search identified a total of 2994 studies (PubMed: 98, Scopus: 1191, Embase: 281, Web of Science:1011, CINAHL:189, Cochrane library:35, PsycINFO:189). After removing the duplicates, the total number of studies left was 2327 and all of these were screened. On screening, 2223 studies failed to meet our inclusion criteria and were removed from the analysis. The full text was obtained for 104 studies. Out of these 104 studies, an additional 92 were excluded and 12 met our inclusion criteria as defined above.(Figure 1)

Risk of bias in the included studies

The quality of cohort studies assessed by NOS, RCTs by Cochrane collaboration's tool, and cross-sectional study by Joanna Briggs Institute Critical Appraisal tool.(Supplementary Table S1, S2, S3)

Characteristics of included studies (Table 1)

Seven studies were designed as prospective cohort studies[23,24,26–30], three were randomized controlled trials (RCT)[20–22], one was a retrospective cohort study[25], and one was a cross-sectional

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study[19]. Two studies were each from Brazil[21,29] and Korea[24,30], and one each from China[23], Japan[25], India[26], UK[20], Germany[27], Indonesia[19], Iran[28], and USA[22]. One study compared QOL in BC patients with NAC and adjuvant therapy[29], and three studies compared QOL in patients with different regimens of NAC[20,26,28]. Two studies were in the same clinical trial registration with different included subjects[24,30]. The QOL-related instruments used were: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30 (EORTC QLQ-C30), the Quality of Life Questionnaire-Breast Cancer Module (QLQ-BR23), The Hospital Anxiety and Depression Scale (HADS), Pittsburgh Sleep Quality Index (PSQI), Multinational Association of Supportive Care in Cancer Antiemesis Tool (MAT), Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast (QOL-ACD-B), Functional Assessment of Cancer Therapy of Breast (FACT-B), Functional Assessment of Cancer Therapy-Taxane (FACT-Taxane), 36-Item Short Form Health Survey questionnaire (SF-36), Body Image Scale (BIS), Rotterdam Symptom Checklist (RSCL), Mood Rating Scale (MRS), Global Distress Scale (GDS), Treatment Side-Effects Questionnaire (TSEQ), Fragebogen erlebter Defizite der Aufmerksamkeit (FEDA), Rosenberg Self Esteem Scale (RSES), Brief Fatigue Inventory (BFI), Center for Epidemiologic Studies–Depression Scale (CES-D).(Table 1)

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Table 1 *Characteristics of included studies*

Author	Year	Country	Study design	Sample size(n)	Age	Regimens of neoadjuvant chemotherapy	Timing of assessment	QOL-related instrument
Ding et al.[23]	2019	China	Prospective cohort study	29	46±8	CEF regimen	Before and after NAC	EORTC QLQ-C30
Lee KM et al.[24]	2017	Korea	Prospective cohort study	134	44.48±7.48	Anthracycline-based regimens along with standard doses of antiemetics	Before and after the first cycle of NAC	PSQI, HADS, MAT
Takada et al.[25]	2018	Japan	Retrospective cohort study	300	55(27-90)	FEC + paclitaxel(HER2-positive: + trastuzumab)	Before and after NAC	QOL-ACD-B
Chellappan[26]	2018	India	Prospective cohort study	50(weekly) vs 50(3-weekly)	Weekly: ≤50 years accounted for 58%, ≥50 years accounted for 42% 3-weekly: ≤50 years accounted for 56%, ≥50 years accounted for 44%	Fifty patients: paclitaxel and doxorubicin (4 course) every three weekly Other fifty: paclitaxel weekly (10 course) along with doxorubicin (4 course) every three weekly	Weekly (Total 10 weeks)	HAM-D, FACT-B, FACT-Taxane
Walker et al.[20]	2011	UK	RCT	41 (Weekly)vs 41(3-weekly)	Weekly: 50.1(27-68); 3-weekly: 48.3(32-70)	Weekly: AC + weekly docetaxel 3-weekly: AC + 3-weekly docetaxel	Before randomisation, every 3 weeks during docetaxel, and 3 weeks after completion of chemotherapy	FACT-B, RSCL, HADS, MRS, GDS, TSEQ
Hermelink et al.[27]	2007	Germany	Prospective cohort study	101	48.6±9.7	Standard chemotherapy or dose-intensified therapy	Before and toward the end of NAC	FEDA, Cognitive Function Scale of EORTC QLQ-C30; HADS
Aprilianto et al.[19]	2021	Indonesia	Cross-sectional study	56	Almost half (33.9%) of respondents were between the ages of 41 to 50 years old.	-	-	A self-made family social support questionnaire, RSES
Rezapour et al.[28]	2018	Iran	Prospective cohort study	50 vs 50	28~70	AC vs PG regimens	At the beginning and end of chemotherapy	EORTC QLQ-C30
Coelho et al.[29]	2017	Brazil	Prospective cohort study	53(NAC) vs 14(adjuvant chemotherapy)	51.3(30~77)	NAC: anthracyclines and taxanes Adjuvant chemotherapy: -	T1: on the day the treatment was started T2: from 40 to 50 days after the first, when the adverse effects began T3: from 40 to 50 days after the second, when the women were already adapted to treatment or the effects were being handled by the team	EORTC QLQ-C30, QLQ-BR23

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Author	Year	Country	Study design	Sample size(n)	Age	Regimens of neoadjuvant chemotherapy	Timing of assessment	QOL-related instrument
Souza et al.[21]	2021	Brazil	RCT	19(IG) vs 15(CG)	44.3±9.2(IG) vs 45.5±8.6(CG)	AC regimen	T0: before the beginning of the first cycle T1: during the second cycles T2: during the third cycles T3: until the end of the third one	EORTC QLQ-C30, HGS
Basen-Engquist et al.[22]	2020	USA	RCT	19(IG) vs 18(CG)	49.6±13.3 (IG) vs 49.2±9.2 (CG)	-	T0: at baseline T1: mid-chemotherapy (3 months) T2: post-chemotherapy (6 months)	SF-36, BIS
Lee S et al.[30]	2021	Korea	Prospective cohort study	193	44.89±7.913	Four sessions of anthracycline and cyclophosphamide, followed by four sessions of docetaxel.	T0: before the first session of NAC T1: before the start of the last session of chemotherapy T2: 6 months after the end of chemotherapy	HADS, FACT-B

-:not reported. *NAC:* neoadjuvant chemotherapy. *CEF*: cyclophosphamide, epirubicin, and 5-fluorouracil. *FEC*: fluorouracil injection, epirubicin, and cyclophosphamide. *AC*: doxorubicin and cyclophosphamide. *PG*: paclitaxel and gemcitabine. *NAST*: neoadjuvant systemic therapy. *IG*: intervention group. *CG:* control group. *EORTC QLQ-C30*: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30. *QLQ-BR23*: the Quality of Life Questionnaire-Breast Cancer Module. *HADS*: The Hospital Anxiety and Depression Scale. *PSQI*: Pittsburgh Sleep Quality Index. *MAT*: Multinational Association of Supportive Care in Cancer Antiemesis Tool. *QOL-ACD-B*: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. *HAM-D*: Hamilton Rating Scale for Depression. *FACT-B*: Functional Assessment of Cancer Therapy of Breast. *FACT-Taxane*: Functional Assessment of Cancer Therapy-Taxane. *SF-36*: 36-Item Short Form Health Survey questionnaire. *BIS*: Body Image Scale. *RSCL*: Rotterdam Symptom Checklist. *MRS*: Mood Rating Scale. *GDS*: Global Distress Scale. *TSEQ*: Treatment Side-Effects Questionnaire. *FEDA*: Fragebogen erlebter Defizite der Aufmerksamkeit (Questionnaire of Experienced Attention Deficits). *RSES*: Rosenberg Self Esteem Scale. *BFI*: Brief Fatigue Inventory. *CES-D*: Center for Epidemiologic Studies-Depression Scale. *HGS*: handgrip strength.

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Quality of life

Physical symptoms

Ten studies evaluated patients' physical aspects with NAC. NAC affects all aspects of a patient's body and 80% of the patients had adverse physical symptoms(Table 2; Figure 2). Three studies evaluated dyspnea[23,28,29] and pain[20,21,25] respectively, two evaluated nausea/vomit[21,24], fatigue[23,28] and cognitive function[27,28] respectively, one evaluated body image[22], and five evaluated other symptoms[21,25,26,28,29]. Lee KM et al.[24] found that 48.5% of patients undergoing NAC experienced chemotherapy-induced nausea and vomiting (CINV), with delayed CINV (prevalence 42.5%) being more common than acute CINV (prevalence 39.6%). Ding et al.[23] reported patients experienced increased dyspnea and fatigue after chemotherapy. Takada et al.[25] found that before NAC, the low QOL group had significantly lower scores on the subscale of Physical symptoms and pain, and the subscale of Dress, sexual aspect, other categories than the high QOL groups (both P<0.001). However, there was no significant difference between the two groups of these subscales after NAC. Hermelink et al.[27] reported during NAC, cognitive function remained stable in most BC patients, but cognitive decline predominated in 27% of patients. It caused by anxiety and depression of chemotherapy.

Different regimens of neoadjuvant chemotherapy

Three studies investigated the changes in physical symptoms with different treatment regimens of NAC. Chellappan[26] showed that there was no difference in physical problems between the weekly regimen of paclitaxel and the 3-weekly regimen of paclitaxel. Rezapour et al.[28] reported that QOL declined in both doxorubicin and cyclophosphamide (AC) and paclitaxel and gemcitabine (PG) groups, but the two were reflected in different symptoms. The AC group had significantly better situation in fatigue, pain, cognitive functioning, and constipation compared to the PG group. Nausea and vomiting, dyspnea, insomnia, and diarrhea were better in the PG group. There was no statistically significant difference in physical performance and physical capability between two groups. Walker et al.[20] reported patients had severe pain in the 3-weekly group compared with the weekly docetaxel group (P=0.034).

Neoadjuvant vs adjuvant chemotherapy

Only one study compared the effects of neoadjuvant and adjuvant chemotherapy on physical symptoms. Coelho et al.[29] observed that QOL was altered with the greater loss for getting NAC compared to adjuvant. Physical function and dyspnea symptoms had a greater impact in the neoadjuvant group comparing to the adjuvant group. It attributed to late diagnoses that provide aggressive treatments.

Intervention in neoadjuvant chemotherapy

Two of these studies intervened in BC patients on NAC. Souza et al.[21] showed that nutritional interventions improved patients' grip strength, positively affected the occurrence of nausea/vomiting and loss of appetite, and reduced the frequency of leukopenia and abdominal pain during NAC. Basen-Engquist et al.[22] reported that weight gain prevention intervention reduced more in waist circumference (P=0.03) and greater improvements in self-reported vitality scores (P=0.03) compared to the control group. The majority of participants reported being satisfied with the

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intervention during chemotherapy (88%).

Table 2 Quality of life of breast cancer patients

Subjects	Themes	Study(n)	Author	Outcomes
Physical symptoms	Body image	1	Basen-Engquist et al.[22]	The difference in body image was not significant between IG and CG.
	Nausea/vomit	2	Lee KM et al.[24]	48.5% of patients undergoing NAC experienced CINV.
			Souza et al.[21]	Nutritional interventions during NAC positively affected the occurrence of nausea/vomiting and loss of appetite.
	Fatigue	2	Ding et al.[23]	Patients experienced increased fatigue after NAC.
			Rezapour et al.[28]	The PG group had increased fatigue than the AC group.
	Cognitive function	2	Hermelink et al.[27]	During NAC, cognitive function remained stable in most patients, but cognitive decline predominated in 27% of patients.
			Rezapour et al.[28]	The PG group had impaired cognitive function than the AC group.
	Dyspnea	3	Ding et al.[23]	Patients experienced increased dyspnea after NAC.
			Coelho et al.[29]	Patients who received NAC had increased dyspnea compared with adjuvant chemotherapy.
			Rezapour et al.[28]	The AC group had increased dyspnea than the PG group.
	Pain	3	Takada et al.[25]	Before NAC, the low QOL group had significantly lower scores on the subscale of Physical symptoms and pain than the high QOL groups(P<0.001). However, there was no significant difference between two groups after NAC.
			Souza et al.[21]	Nutritional interventions reduced the frequency of abdominal pain during NAC.
			Walker et al.[20]	The pain was significantly greater in the 3-weekly group compared with the weekly group(P=0.034).
	Other symptoms	5	Chellappan et al.[26]	There was no difference in physical problems between the weekly regimen of paclitaxel and the 3-weekly regimen of paclitaxel.
			Takada et al.[25]	Before NAC, the low QOL group had significantly lower scores on the subscale of Dress, sexual aspect, other categories than the high QOL groups(P<0.001). However, there was no significant difference between two groups after NAC.
			Rezapour et al.[28]	The AC group had significantly better situation in constipation compared to the PG group. Insomnia and diarrhea were better in the PG group.
			Coelho et al.[29]	Physical function had a greater impact in the neoadjuvant group comparing to the adjuvant group.
			Souza et al.[21]	Nutritional interventions improved patients' grip strength, positively affected the loss of appetite, and reduced the frequency of leukopenia during NAC.
Psychological Effects	Role function	2	Rezapour et al.[28]	The AC group had a better role function compared to the PG group.
			Souza et al.[21]	Nutritional interventions were able to improve the role function.
	Anxiety/depression	3	Lee S et al.[30]	The depression in BC patients were most severe during NAC compared to before and after NAC. Anxiety was most severe before NAC, but gradually improved throughout the treatment session.
			Lee KM et al.[24]	40 patients were anxious and 49 were depressed. In the univariate analyses, overall CINV was significantly associated with anxiety (P=0.036).
			Walker et al.[20]	Anxiety and depression were not statistically significant between the weekly group and the 3-weekly group.

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Subjects	Themes	Study(n)	Author	Outcomes
	Emotional function	4	Chellappan et al.[26]	Three weekly patients had more problems regarding emotional well-being than weekly patients. But the observed difference in scores of depression among the two groups was not statistically significant.
			Rezapour et al.[28]	The PG group had better emotional functioning compared to the AC group.
			Coelho et al.[29]	The emotional function was significantly affected in adjuvant chemotherapy and NAC, but there was no difference in the effects of the two treatments.
			Basen-Engquist et al. [22]	Differences in role emotional were not significant between IG and CG.
Support	Family support	1	Aprilianto et al.[19]	36 respondents (64.3%) were in a good category of family social support. The self-esteem of almost half of the BC patients (27 respondents or 48.2%) was is in a moderate level category.
	Social function	4	Chellappan et al.[26]	Weekly group had a high level of social well-being than 3-weekly group.
			Walker et al.[20]	There was no significant difference in the social dimension between weekly group and 3-weekly group.
			Basen-Engquist et al.[22]	Differences in social function of IG and CG groups were not significant.
			Rezapour et al.[28]	The AC group had better social function compared to the PG group.

NAC: neoadjuvant chemotherapy. *QOL*: quality of life. *AC*: doxorubicin and cyclophosphamide. *PG*: paclitaxel and gemcitabine. *CINV*: chemotherapy-induced nausea and vomiting. *IG*: intervention group. *CG:* control group.

Psychological Effects

Eight studies evaluated psychological effects during NAC and 52% of patients get negative effects on psychological aspects (Table 2; Figure 2). Four studies evaluated emotional function[22,26,28,29], three studies evaluated anxiety/depression[20,24,30], and two evaluated role function[21,28]. Lee S et al.[30] reported that the depression in BC patients were most severe during NAC compared to before and after NAC. Anxiety was most severe before NAC, but gradually improved throughout the treatment session. Lee KM et al.[24] reported that overall CINV was significantly associated with anxiety.

Different regimens of neoadjuvant chemotherapy

Three studies investigated the changes in psychological effects with different regimens of NAC. Chellappan et al.[26] studied NAC with different regimens and showed that patients who were treated once a week had a high level of emotional well-being than those who were treated three times a week. But depression among the two groups was not statistically significant. Meanwhile Walker et al.[20] found anxiety and depression were not statistically significant between the weekly group and the 3-weekly group. Rezapour et al.[28] reported that the AC group had a high level of role functioning than the PG group. The emotional functioning was better in the PG group compared to the AC group.

Neoadjuvant vs adjuvant chemotherapy

One study investigated the psychological effects between NAC and adjuvant chemotherapy. Coelho et al.[29] reported that emotional function was significantly affected in adjuvant chemotherapy and NAC, but there was no difference in the effects of the two treatments.

Intervention in neoadjuvant chemotherapy

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Two studies intervened in BC patients on NAC. Souza et al.[21] reported that nutritional interventions were able to improve the role function. Basen-Engquist et al.[22] reported that weight gain prevention intervention didn't change mental health or role emotion.

Support

Five studies evaluated support from society or family and 55% of patients did not receive adequate support (Table 2; Figure 2). Aprilianto et al.[19] reported that 64.3% of the respondents reported that social support for the patient's family was mostly in the good category, and 48.2% of the BC patients had a moderate level of self-esteem. There was a strong positive correlation between family social support and patient self-esteem.

Different regimens of neoadjuvant chemotherapy

Three studies evaluated support between different regimens of NAC. Chellappan et al.[26] reported that weekly group had a high level of social well-being than 3-weekly group (P=0.007). Rezapour et al.[28] reported that the AC group had significantly better social function compared to the PG group. Walker et al.[20] showed that the weekly group and the 3-weekly group had no significant difference in the social dimension.

Intervention in neoadjuvant chemotherapy

Basen-Engquist et al.[22] reported weight gain prevention intervention couldn't change social function of patients.

Quality of life-related predictors

Three studies have extrapolated QOL-related predictors of BC patients during NAC: chronotype, QOL before NAC, and depression[24,25,30]. Lee KM et al.[24] suggests late chronotype may be a predictor of CINV. BC patients with late chronotype have an increased risk of CINV during NAC. In another study, Lee S et al.[30] indicated that depression during NAC was a mediator in the relationship between resilience and health-related QOL in BC patients, and QOL after NAC can be improved by screening and intervening for depression during NAC in BC patients. Meanwhile Takada et al.[25] reported that high QOL before chemotherapy was an independent factor in overall survival. (Table 3)

Table 3 *Quality of life-related predictors*

Study	Sample size(n)	Study design	Instrument	Predictors
Lee KM et al.[24]	134	Prospective cohort study	PSQI, HADS, MAT	Late chronotype may be a predictor of CINV.
Takada et al.[25]	300	Retrospective cohort study	QOL-ACD-B	The level of QOL before chemotherapy was an independent factor in overall survival.
Lee S et al.[30]	193	Prospective cohort study	HADS, FACT-B	Depression was a mediator in the relationship between resilience and health-related QOL.

QOL-ACD-B: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. *FACT-B*: Functional Assessment of Cancer Therapy of Breast. *HADS*: The Hospital Anxiety and Depression Scale. *PSQI*: Pittsburgh Sleep Quality Index. *MAT*: Multinational Association of Supportive Care in Cancer Antiemesis Tool. *CINV*: chemotherapy-induced nausea and vomiting.

Discussion

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In recent years, NAC has been widely used in the treatment of BC, and its efficiency in early-stage BC is as high as 70%~90%, and has improved the success rate of breast-conserving surgery and the long-term prognosis of patients in complete pathological remission[31,32]. However, it also has significant side effects on various organs of patients. Patients with BC after NAC require further surgical treatment. Chemotherapy-related symptoms and poor QOL inevitably affect the patient's surgical tolerance and it may delay the patient's surgical treatment in severe cases. Therefore, information about expected QOL will play an important role in the decision-making process of patients and clinicians considering NAC, so this study was conducted to systematically evaluate the QOL of NAC patients.

The results of this systematic review showed no significant change in overall QOL before and after NAC in BC patients. However, during chemotherapy, patients' QOL decreased, patients experienced increased pain, nausea and vomiting, dyspnea, and fatigue[23–25,28,29], and nausea remained unimproved despite the use of an antiemetic regimen that relieved vomiting[24]. Patients' QOL decreases during chemotherapy due to the toxic effects of chemotherapy drugs, and after chemotherapy, some of the toxic effects diminish and patients' QOL improves. Emotional problems such as anxiety and depression are also aggravated during chemotherapy, but they all resolve at the end of chemotherapy[29,30]. Emotional problems during NAC showed a significant association with resilience before NAC. These findings are largely consistent with previous studies on QOL during NAC for other cancers, such as esophageal cancer[33], rectal cancer[34], pancreatic cancer[35] and muscle-invasive bladder cancer[36]. Different regimens of NAC have different aspects of impact on patients' QOL. The regimen of doxorubicin, cyclophosphamide, and docetaxel weekly had less distressing side effects compared to the 3-weekly group[20], and another regimen of paclitaxel and doxorubicin had fewer social well-being compared to weekly group[26]. Comparisons of AC and PG regimens showed that PG had fewer physical adverse effects and AC had better psychological effects[28]. Only one study of included studies compared the impact of NAC to adjuvant chemotherapy on BC patients' QOL[29], both regimens could affect patients emotionally. However, NAC was more damaging in the terms of physical, cognitive and personal performance due to more toxic of the drugs in NAC. QOL can be improved in various ways, including appropriate interventions and enhanced social or family support. Past studies reported that nutritional intervention can improve the symptoms of nausea, vomiting, and loss of appetite in BC patients, and exercise can reduce fatigue and improve the physical quality of BC patients[21,37,38]. BC patients with high levels of social support have lower severity of chemotherapy-related symptoms[39]. In addition, interventions can be based on QOL predictors of BC, such as chronotype, the level of QOL before chemotherapy, and depression.

The limitations of this study are mainly in the low inclusion of literature on some sub-topics and insufficient evidence, especially the contrast between NAC and adjuvant chemotherapy.

More original studies of high quality are needed. Potential area for further investigations provided by this systematic review is about QOL in the perioperative and long-term postoperative periods after NAC. It is important to improve long-term survival prognosis about NAC but also to focus on and improve QOL during and after NAC.

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Conclusion

The results of this systematic review showed that BC patients' QOL was significantly lower during NAC. There were no significant change in overall QOL before and after NAC. And this review found that the influences are mainly reflected in physical symptoms(80%), psychological effects(52%) and support from society or family (55%). QOL could be improved with appropriate interventions. Factors that affect QOL include chronotype, QOL before NAC, and depression can affect the QOL. More original research is needed in the future to understand the profile and predictors of QOL in patients on neoadjuvant therapy, which will help clinicians and patients make better decisions and deal with NAC-related issues.

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Footnotes

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Author Contributions: All the authors contributed to study design. YQ Z and XQ Z screened abstracts and full-text articles for inclusion. YQ Z and LC C appraised study quality. Any disagreements or uncertainties during the screening and quality appraisal process were referred to YL S. YQ Z drafted the manuscript, which was reviewed, edited and approved by YL S.

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Competing interests: None declared.

Patient consent: No.

Data availability statement: Data are available upon reasonable request. The data are available upon request from the corresponding author (2187009@zju.edu.cn)

Ethics Approval: This study was a systematic review, so we were not provided the ethics approval.

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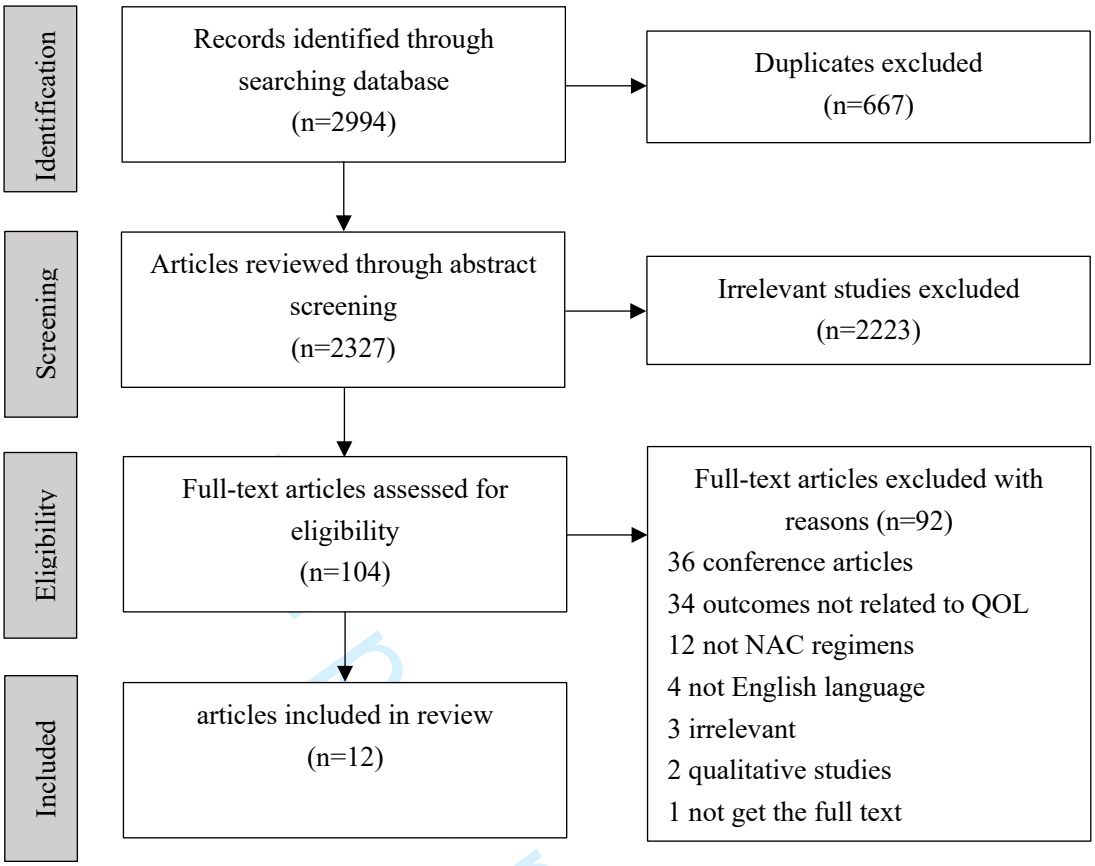


Figure 1 Flow chart for systematic review methodology as per PRISMA guidelines

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Figure 2 Summary results of the proportion of each negative impact on quality of life

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Supplementary:search strategies

Web of science Search strategy

#4 #1AND #2 AND #3
#3 TS=("Neoadjuvant chemotherapy")
#2 (((((((((TS=("Quality of Life")) OR TS=("life quality")) OR TS=(QOL)) OR TS=(HRQOL)) OR TS=(Support)) OR TS=(depress*)) OR TS=(anxiety)) OR TS=(sexual*)) OR TS=(psychological)) OR TS=(mental)
#1 (((TS=("Breast Neoplasms")) OR TS=("breast tumor")) OR TS=("breast cancer")) OR TS=("breast carcinoma")

Pubmed Search strategy

#8 #4 AND #5 AND #7
#7 #6 OR #2
#6 "psychological"[Title/Abstract] OR "mental"[Title/Abstract] OR "depress*"[Title/Abstract] OR "HRQOL"[Title/Abstract] OR "life quality"[Title/Abstract]
#5 #1OR #3
#4 "Neoadjuvant Therapy"[MeSH Terms]
#3 "breast tumor"[Title/Abstract] OR "breast cancer"[Title/Abstract] OR "breast carcinoma"[Title/Abstract]
#2 "Quality of Life"[MeSH Terms] OR "Social Support"[MeSH Terms] OR "Depression"[MeSH Terms] OR "Anxiety"[MeSH Terms] OR "Sexuality"[MeSH Terms]
#1 "Breast Neoplasms"[MeSH Terms]

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Embase Search strategy

#9 #6 AND #7 AND #8

#8 'neoadjuvant chemotherapy'/exp

#7 'breast tumor'/exp

#6 #1 OR #2 OR #3 OR #4 OR #5

#5 'sexuality'/exp

#4 'depression'/exp

#3 'anxiety'/exp

#2 'quality of life'/exp

#1 'social support'/exp

Cochrane Library Search strategy

#9 #6 AND #7 AND #8

#8 #1 OR #2 OR #3 OR #4 OR #5

#7 MeSH descriptor:[Neoadjuvant Therapy] explode all trees

#6 MeSH descriptor:[Breast neoplasms] explode all trees

#5 MeSH descriptor:[Sexuality] explode all trees

#4 MeSH descriptor:[Depression] explode all trees

#3 MeSH descriptor:[Anxiety] explode all trees

#2 MeSH descriptor:[Social Support] explode all trees

#1 MeSH descriptor:[Quality of life] explode all trees

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Scopus/PsyCINFO/CINAHL Search strategy

#4 #1 AND #2 AND #3

#3 "psychological" OR "mental" OR "anxiety" OR "depress*" OR "quality of life" OR "HRQOL" OR "QOL" OR "life quality" OR "sexual*" OR "support"

#2 "neoadjuvant chemotherapy"

#1 "breast cancer" OR "breast tumor" OR "breast carcinoma" OR "breast neoplasms"

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Supplementary Table S1 *Quality of RCT studies as assessed by Cochrane collaboration's tool for assessing risk of bias*

Study	Random Sequence Generation (selection bias)	Allocation Concealment (selection bias)	Blinding of Participants and Personnel (performance bias)	Blinding of Outcome Assessors (detection bias)	Completeness of Reporting Outcome Data (attrition bias)	Selective Outcome Reporting (reporting bias)	other bias
Walker et al.[20]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Unclear risk
Souza et al.[21]	Low risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk	Low risk
Basen-Engquist et al.[22]	Low risk	Unclear risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk

Supplementary Table S2 *Quality of cohort studies as assessed by Newcastle Ottawa Score(NOS)*

Study	Representativeness of exposed cohort	Selection non-exposed cohort	Ascertainment of exposure	Outcome of interest	Comparability	Assessment of outcome	Length of follow up	Adequacy of follow up	Total score
Ding et al.[23]	*	-	*	-	-	*	*	*	5
Lee KM et al.[24]	*	-	*	-	-	*	*	*	5
Takada et al.[25]	*	*	*	-	**	*	*	*	8
Chellappan[26]	*	*	*	-	**	*	*	*	8
Hermelink et al.[27]	*	-	*	-	-	*	*	*	5
Rezapour et al.[28]	*	*	*	-	*	*	*	*	7
Coelho et al.[29]	*	*	*	-	-	*	*	*	6
Lee, S et al.[30]	*	-	*	-	*	*	*	*	6

Supplementary Table S3 *Quality of cross-sectional study as assessed by JBI Critical Appraisal Checklist*

Items	Aprilianto et al.[19]
1. Were the criteria for inclusion in the sample clearly defined?	N
2. Were the study subjects and the setting described in detail?	Y
3. Was the exposure measured in a valid and reliable way?	Y
4. Were objective, standard criteria used for measurement of the condition?	Y
5. Were confounding factors identified?	Y
6. Were strategies to deal with confounding factors stated?	N
7. Were the outcomes measured in a valid and reliable way?	Y
8. Was appropriate statistical analysis used?	Y
Overall appraisal	Include

Y: yes. *N*: no.

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	Page 1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	Page 1
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	Page 2
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	Page 2
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	Page 3
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	Page 2
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	Page 2
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they	Page 3

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		worked independently, and, if applicable, details of automation tools used in the process	
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	Page 3
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	Page 3
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	Page 3
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	Page 3

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Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	Page 3-4
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	Page 3
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	Page 3
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	Page 3
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	n/a.This review could not conduct by meta-analysis.
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	n/a.This review could not conduct by meta-analysis.

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Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	Page 3
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	n/a.This review could not conduct by meta-analysis.
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	Page 3
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	Page 3-4
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	Page 3-4
Study characteristics	#17	Cite each included study and present its	Page 5

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		characteristics	
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	Page 3
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	Page 5
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	Page 5
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	Page 7-11
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	n/a.This review could not conduct by meta-analysis.
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	n/a.This review could not conduct by

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			meta-analysis.
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	n/a.This review could not conduct by meta-analysis.
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	n/a.This review could not conduct by meta-analysis.
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	Page 11-12
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	Page 12
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	Page 12
Implications	#23d	Discuss implications of the results for practice, policy, and future research	Page 12
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	n/a.Registration is still under review.

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Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	n/a.Registration is still under review.
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	n/a.
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	Page 12
Competing interests	#26	Declare any competing interests of review authors	Page 12
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	Page 13

Notes:

- 13e: n/a.This review could not conduct by meta-analysis.
- 13f: n/a.This review could not conduct by meta-analysis.
- 15: n/a.This review could not conduct by meta-analysis.
- 20c: n/a.This review could not conduct by meta-analysis.
- 20d: n/a.This review could not conduct by meta-analysis.
- 21: n/a.This review could not conduct by meta-analysis.

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- 22: n/a.This review could not conduct by meta-analysis.
- 24a: n/a.Registration is still under review.
- 24b: n/a.Registration is still under review. The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 11. February 2022 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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Quality of life in breast cancer patients with neoadjuvant chemotherapy: a systematic review

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Abstract

Objectives Neoadjuvant chemotherapy (NAC) shrinks tumors and improves survival in breast cancer (BC) patients. But it can also have an impact on patients' quality of life (QOL). Therefore, it is important to assess the QOL of BC patients during and after NAC. The aims of this systematic review were (1) to assess the impact of NAC on BC patients' QOL; (2) to compare the different regimens of NAC on BC patients' QOL; (3) to compare NAC versus adjuvant chemotherapy on BC patients' QOL; (4) to identify predictors of QOL on BC patients receiving NAC.

Methods We performed a systematic review of literature using the PRISMA guidelines. Cinahl, Embase, Pubmed, Scopus, Web of Science, Cochrane library, and PsycINFO databases were searched. The final search was in December 2021. The inclusion criteria were as follows: (1) adult patients (18 years older); (2) BC patients receiving NAC; (3) outcome measures related to QOL, including QOL, psychological impact, support from various sources, and so on; (4) published in the English language. The exclusion criteria were as follows: (1) data could not be extracted; (2) duplicates or overlapping participants; (3) not original research, such as commentaries, editorials, or reviews;(4) full text not available;(5) qualitative study. The risk of bias in individual studies was evaluated with Cochrane collaboration's tool for assessing risk bias, Newcastle Ottawa Score(NOS), or Joanna Briggs Institute Critical Appraisal tool. This systematic review performs narrative synthesis based on several different themes.

Results The initial search resulted in 2994 studies; 12 of these studies fulfilled our inclusion criteria. There was no significant difference in the QOL of BC before and after NAC, but patients experienced adverse reactions and depression during chemotherapy. Different regimens of NAC have different effects on patients' QOL. Patients with NAC had more severe physical discomfort than those with adjuvant chemotherapy. However, BC patients' QOL can be improved by intervening on social or family support, and these predictors including chronotype, QOL before NAC, and depression.

Conclusion More original research is needed in the future to understand the profile and predictors of QOL in BC patients on NAC, which will help clinicians and patients make decisions and deal with NAC-related issues.

Other This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Article Summary

Strengths and limitations of this study

This systematic review was designed and reported as PRISMA guidelines.

This review studied the impact and predictors of neoadjuvant chemotherapy on the quality of life of breast cancer patients.

The electronic search will only include studies published in English that could limit the inclusion of studies.

Keywords Breast cancer; Neoadjuvant chemotherapy; Quality of life; Systematic review

Introduction

Breast cancer (BC) is the most common disease in women worldwide, with the prevalence now surpassing lung cancer as the leading global cancer incidence in women in 2020, with 11.7% of all new cases (2.3 million); and it is the fifth leading cause of cancer deaths worldwide, accounting for 6.9%.[1] Treatment of BC usually consists of two parts: local surgical treatment and systemic treatment; systemic treatment can precede surgical treatment (neoadjuvant) or follow it (adjuvant).[2] Neoadjuvant chemotherapy (NAC) is used in patients with locally advanced BC or inoperable BC to improve the likelihood of breast-conserving surgery by reducing the size of the tumor in the past. Current studies have shown that NAC can also be used in the treatment of patients with early-stage operable BC to improve cosmetic outcomes and reduce postoperative complications such as lymphedema and prolong survival.[3–5] NAC is associated with improved survival compared with adjuvant chemotherapy in patients with triple-negative BC only after complete pathologic response,[6] and the same effect has been observed in other cancers like bladder cancer and colon cancer.[7,8] NAC is widely used, while more and more scholars are studying the survival outcomes of patients after NAC, such as pathologic complete response, disease-free survival and overall survival.[4,9]

However, the goals of cancer treatment should include not only survival outcomes but also quality of life (QOL), which is an important outcome measure in clinical investigations and survival studies in BC.[10–12] Now the treatment regimens of BC are numerous and effective, but the side effects associated with different treatment modalities and the impact on QOL are different. So optimal stratification and ranking of treatment strategies should focus on individual risk profiles as well as QOL.[13] Meanwhile, QOL is a major factor in treatment decisions for advanced or metastatic disease.[14] It is worthwhile to consider the choice of an effective, least toxic anticancer treatment before implementing definitive therapy. Therefore, studying the QOL of NAC patients and comparing it to other treatments can help patients and clinicians make decisions.

The aims of this systematic review were (1) to assess the impact of NAC on BC patients' QOL; (2) to compare the different regimens of NAC on BC patients' QOL; (3) to compare NAC versus adjuvant chemotherapy on BC patients' QOL; (4) to identify predictors of QOL on BC patients receiving NAC.

Methods

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We primarily followed the Joanna Briggs Institute guidelines[15] on systematic reviews and we used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[16] (Figure 1)

Search strategy

A systematic search of the following databases was performed: CINAHL, Embase, Pubmed, Scopus, Web of Science, Cochrane library, and PsycINFO. The final search was in December 2021. Three categories of terms were searched: (1)breast cancer, (2) neoadjuvant chemotherapy, (3) quality of life. In PubMed and Cochrane library, Medical Subject Headings (MeSH) were used (breast neoplasms, neoadjuvant therapy, quality of life, sexuality, anxiety, depression, social support). In Embase, Emtree terms were exploded (breast tumor, neoadjuvant chemotherapy, quality of life, sexuality, anxiety, depression, social support). In Scopus, Web of Science, CINAHL, and PsycINFO, only keywords were used. (Supplementary material)

Data analysis

The inclusion criteria were as follows: (1) adult patients (18 years older); (2) BC patients receiving NAC; (3) outcome measures related to QOL, including QOL, psychological impact, support from various sources and so on; (4) published in the English language. The exclusion criteria were as follows: (1) data could not be extracted; (2) duplicates or overlapping participants; (3) not original research, such as commentaries, editorials, or reviews;(4) full text not available;(5) qualitative study. Two independent researchers (YQ Z and XQ Z) screened for potentially relevant studies by reviewing abstracts. The full texts were then screened further according to the inclusion and exclusion criteria. Any uncertainty was solved by a third researcher(YL S). Two researchers (YQ Z and XQ Z) independently extracted and recorded the following information from the enrolled studies: author, published year, country, study design, sample size, age, regimens of NAC, timing of assessment, instrument, outcomes. The risk of bias in individual studies was evaluated by 2 independent researchers (YQ Z and LC C) in accordance with Cochrane collaboration's tool for assessing risk bias,[17] or Newcastle Ottawa Score(NOS)[18] or Joanna Briggs Institute Critical Appraisal tool.[19] Any disagreement was resolved by a third researcher(YL S).

Patient and public involvement

Patients or the public were not involved in the conceptualisation or carrying out of this research.

Results

Flow of included studies

An initial search identified a total of 2994 studies (PubMed: 98, Scopus: 1191, Embase: 281, Web of Science:1011, CINAHL:189, Cochrane library:35, PsycINFO:189). After removing the

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duplicates, the total number of studies left was 2327 and all of these were screened. On screening, 2223 studies failed to meet our inclusion criteria and were removed from the analysis. The full text was obtained for 104 studies. Out of these 104 studies, an additional 92 were excluded and 12 met our inclusion criteria as defined above. (Figure 1)

Risk of bias in the included studies

The quality of cohort studies was assessed by NOS, RCTs by Cochrane collaboration's tool, and cross-sectional study by Joanna Briggs Institute Critical Appraisal tool. (Supplementary Table S1, S2, S3)

Characteristics of included studies (Table 1)

Seven studies were designed as prospective cohort studies,[20–25] three were randomized controlled trials (RCT),[26–28] one was a retrospective cohort study,[29] and one was a cross-sectional study.[30] Two studies were each from Brazil[25,27] and Korea,[21,31] and one each from China,[20] Japan,[29] India,[22] UK,[26] Germany,[23] Indonesia,[30] Iran,[24] and USA.[28] One study compared QOL in BC patients with NAC and adjuvant therapy,[25] and three studies compared QOL in patients with different regimens of NAC.[22,24,26] Two studies were in the same clinical trial registration with different included subjects.[21,31] The QOL-related instruments used were: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30 (EORTC QLQ-C30), the Quality of Life Questionnaire-Breast Cancer Module (QLQ-BR23), The Hospital Anxiety and Depression Scale (HADS), Pittsburgh Sleep Quality Index (PSQI), Multinational Association of Supportive Care in Cancer Antiemesis Tool (MAT), Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast (QOL-ACD-B), Functional Assessment of Cancer Therapy of Breast (FACT-B), Functional Assessment of Cancer Therapy-Taxane (FACT-Taxane), 36-Item Short Form Health Survey questionnaire (SF-36), Body Image Scale (BIS), Rotterdam Symptom Checklist (RSCL), Mood Rating Scale (MRS), Global Distress Scale (GDS), Treatment Side-Effects Questionnaire (TSEQ), Fragebogen erlebter Defizite der Aufmerksamkeit (FEDA), Rosenberg Self Esteem Scale (RSES), Brief Fatigue Inventory (BFI), Center for Epidemiologic Studies–Depression Scale (CES-D). (Table 1)

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Table 1 *Characteristics of included studies*

Author	Year	Country	Study design	Sample size(n)	Age	Regimens of neoadjuvant chemotherapy	Timing of assessment	QOL-related instrument
Ding et al.[20]	2019	China	Prospective cohort study	29	46±8	CEF regimen	Before and after NAC	EORTC QLQ-C30
Lee KM et al.[21]	2017	Korea	Prospective cohort study	134	44.48±7.48	Anthracycline-based regimens along with standard doses of antiemetics	Before and after the first cycle of NAC	PSQI, HADS, MAT
Takada et al.[29]	2018	Japan	Retrospective cohort study	300	55(27-90)	FEC + paclitaxel(HER2-positive: + trastuzumab)	Before and after NAC	QOL-ACD-B
Chellappan[22]	2018	India	Prospective cohort study	50(weekly) vs 50(3-weekly)	Weekly: ≤50 years accounted for 58%, ≥50 years accounted for 42% 3-weekly: ≤50 years accounted for 56%, ≥50 years accounted for 44%	Fifty patients: paclitaxel and doxorubicin (4 course) every three weekly Other fifty: paclitaxel weekly (10 course) along with doxorubicin (4 course) every three weekly	Weekly (Total 10 weeks)	HAM-D, FACT-B, FACT-Taxane
Walker et al.[26]	2011	UK	RCT	41 (Weekly)vs 41(3-weekly)	Weekly: 50.1(27-68); 3-weekly: 48.3(32-70)	Weekly: AC + weekly docetaxel 3-weekly: AC + 3-weekly docetaxel	Before randomisation, every 3 weeks during docetaxel, and 3 weeks after completion of chemotherapy	FACT-B, RSCL, HADS, MRS, GDS, TSEQ
Hermelink et al.[23]	2007	Germany	Prospective cohort study	101	48.6±9.7	Standard chemotherapy or dose-intensified therapy	Before and toward the end of NAC	FEDA, Cognitive Function Scale of EORTC QLQ-C30; HADS
Aprilianto et al.[30]	2021	Indonesia	Cross-sectional study	56	Almost half (33.9%) of respondents were between the ages of 41 to 50 years old.	-	-	A self-made family social support questionnaire, RSES
Rezapour et al.[24]	2018	Iran	Prospective cohort study	50 vs 50	28~70	AC vs PG regimens	At the beginning and end of chemotherapy	EORTC QLQ-C30
Coelho et al.[25]	2017	Brazil	Prospective cohort study	53(NAC) vs 14(adjuvant chemotherapy)	51.3(30~77)	NAC: anthracyclines and taxanes Adjuvant chemotherapy: -	T1: on the day the treatment was started T2: from 40 to 50 days after the first, when the adverse effects began T3: from 40 to 50 days after the second, when the women were already adapted to treatment or the effects were being handled by the team	EORTC QLQ-C30, QLQ-BR23
Souza et al.[27]	2021	Brazil	RCT	19(IG) vs 15(CG)	44.3±9.2(IG) vs	AC regimen	T0: before the beginning of the first	EORTC QLQ-C30,

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Author	Year	Country	Study design	Sample size(n)	Age	Regimens of neoadjuvant chemotherapy	Timing of assessment	QOL-related instrument
					45.5±8.6(CG)		cycle T1: during the second cycles T2: during the third cycles T3: until the end of the third one	HGS
Basen-Engquist et al.[28]	2020	USA	RCT	19(IG) vs 18(CG)	49.6±13.3 (IG) vs 49.2±9.2 (CG)	-	T0: at baseline T1: mid-chemotherapy (3 months) T2: post-chemotherapy (6 months)	SF-36, BIS
Lee S et al.[31]	2021	Korea	Prospective cohort study	193	44.89±7.913	Four sessions of anthracycline and cyclophosphamide, followed by four sessions of docetaxel.	T0: before the first session of NAC T1: before the start of the last session of chemotherapy T2: 6 months after the end of chemotherapy	HADS, FACT-B

-: not reported. *NAC:* neoadjuvant chemotherapy. *CEF*: cyclophosphamide, epirubicin, and 5-fluorouracil. *FEC*: fluorouracil injection, epirubicin, and cyclophosphamide. *AC*: doxorubicin and cyclophosphamide. *PG*: paclitaxel and gemcitabine. *NAST*: neoadjuvant systemic therapy. *IG*: intervention group. *CG:* control group. *EORTC QLQ-C30*: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30. *QLQ-BR23*: the Quality of Life Questionnaire-Breast Cancer Module. *HADS*: The Hospital Anxiety and Depression Scale. *PSQI*: Pittsburgh Sleep Quality Index. *MAT*: Multinational Association of Supportive Care in Cancer Antiemesis Tool. *QOL-ACD-B*: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. *HAM-D*: Hamilton Rating Scale for Depression. *FACT-B*: Functional Assessment of Cancer Therapy of Breast. *FACT-Taxane*: Functional Assessment of Cancer Therapy-Taxane. *SF-36*: 36-Item Short Form Health Survey questionnaire. *BIS*: Body Image Scale. *RSCL*: Rotterdam Symptom Checklist. *MRS*: Mood Rating Scale. *GDS*: Global Distress Scale. *TSEQ*: Treatment Side-Effects Questionnaire. *FEDA*: Fragebogen erlebter Defizite der Aufmerksamkeit (Questionnaire of Experienced Attention Deficits). *RSES*: Rosenberg Self Esteem Scale. *BFI*: Brief Fatigue Inventory. *CES-D*: Center for Epidemiologic Studies-Depression Scale. *HGS*: handgrip strength.

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Quality of life

Physical symptoms

Ten studies evaluated patients' physical aspects with NAC. NAC affects all aspects of a patient's body and 80% of the patients had adverse physical symptoms (Table 2; Figure 2). Three studies evaluated dyspnea[20,24,25] and pain[26,27,29] respectively, two evaluated nausea/vomit,[21,27] fatigue[20,24] and cognitive function[23,24] respectively, one evaluated body image,[28] and five evaluated other symptoms.[22,24,25,27,29] Lee KM et al.[21] found that 48.5% of patients undergoing NAC experienced chemotherapy-induced nausea and vomiting (CINV), with delayed CINV (prevalence 42.5%) being more common than acute CINV (prevalence 39.6%). Ding et al.[20] reported patients experienced increased dyspnea and fatigue after chemotherapy. Takada et al.[29] found that before NAC, the low QOL group had significantly lower scores on the subscale of Physical symptoms and pain, and the subscale of Dress, sexual aspect, other categories than the high QOL groups (both P<0.001). However, there was no significant difference between the two groups of these subscales after NAC. Hermelink et al.[23] reported during NAC, cognitive function remained stable in most BC patients, but cognitive decline predominated in 27% of patients. It is caused by anxiety and depression from chemotherapy.

Different regimens of neoadjuvant chemotherapy

Three studies investigated the changes in physical symptoms with different treatment regimens of NAC. Chellappan[22] showed that there was no difference in physical problems between the weekly regimen of paclitaxel and the 3-weekly regimen of paclitaxel. Rezapour et al.[24] reported that QOL declined in both doxorubicin and cyclophosphamide (AC) and paclitaxel and gemcitabine (PG) groups, but the two were reflected in different symptoms. The AC group had a significantly better situation in fatigue, pain, cognitive functioning, and constipation compared to the PG group. Nausea and vomiting, dyspnea, insomnia, and diarrhea were better in the PG group. There was no statistically significant difference in physical performance and physical capability between the two groups. Walker et al.[26] reported patients had severe pain in the 3-weekly group compared with the weekly docetaxel group (P=0.034).

Neoadjuvant vs adjuvant chemotherapy

Only one study compared the effects of neoadjuvant and adjuvant chemotherapy on physical symptoms. Coelho et al.[25] observed that QOL was altered with the greater loss for getting NAC compared to adjuvant. Physical function and dyspnea symptoms had a greater impact in the neoadjuvant group compared to the adjuvant group. It is attributed to late diagnoses that provide aggressive treatments.

Intervention in neoadjuvant chemotherapy

Two of these studies intervened in BC patients on NAC. Souza et al.[27] showed that nutritional interventions improved patients' grip strength, positively affected the occurrence of nausea/vomiting and loss of appetite, and reduced the frequency of leukopenia and abdominal pain during NAC. Basen-Engquist et al.[28] reported that weight gain prevention intervention reduced more in waist circumference (P=0.03) and greater improvements in self-reported vitality scores (P=0.03) compared to the control group. The majority of participants reported being satisfied with the

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intervention during chemotherapy (88%).

Table 2 Quality of life of breast cancer patients

Subjects	Themes	Study(n)	Author	Outcomes
Physical symptoms	Body image	1	Basen-Engquist et al.[28]	The difference in body image was not significant between IG and CG.
	Nausea/vomit	2	Lee KM et al.[21]	48.5% of patients undergoing NAC experienced CINV.
			Souza et al.[27]	Nutritional interventions during NAC positively affected the occurrence of nausea/vomiting and loss of appetite.
	Fatigue	2	Ding et al.[20]	Patients experienced increased fatigue after NAC.
			Rezapour et al.[24]	The PG group had increased fatigue than the AC group.
	Cognitive function	2	Hermelink et al.[23]	During NAC, cognitive function remained stable in most patients, but cognitive decline predominated in 27% of patients.
			Rezapour et al.[24]	The PG group had impaired cognitive function than the AC group.
	Dyspnea	3	Ding et al.[20]	Patients experienced increased dyspnea after NAC.
			Coelho et al.[25]	Patients who received NAC had increased dyspnea compared with adjuvant chemotherapy.
			Rezapour et al.[24]	The AC group had increased dyspnea than the PG group.
	Pain	3	Takada et al.[29]	Before NAC, the low QOL group had significantly lower scores on the subscale of Physical symptoms and pain than the high QOL groups(P<0.001). However, there was no significant difference between the two groups after NAC.
			Souza et al.[27]	Nutritional interventions reduced the frequency of abdominal pain during NAC.
			Walker et al.[26]	The pain was significantly greater in the 3-weekly group compared with the weekly group(P=0.034).
	Other symptoms	5	Chellappan et al.[22]	There was no difference in physical problems between the weekly regimen of paclitaxel and the 3-weekly regimen of paclitaxel.
			Takada et al.[29]	Before NAC, the low QOL group had significantly lower scores on the subscale of Dress, sexual aspect, other categories than the high QOL groups(P<0.001). However, there was no significant difference between the two groups after NAC.
			Rezapour et al.[24]	The AC group had a significantly better situation in constipation compared to the PG group. Insomnia and diarrhea were better in the PG group.
			Coelho et al.[25]	The physical function had a greater impact in the neoadjuvant group compared to the adjuvant group.
			Souza et al.[27]	Nutritional interventions improved patients' grip strength, positively affected the loss of appetite, and reduced the frequency of leukopenia during NAC.
Psychological Effects	Role function	2	Rezapour et al.[24]	The AC group had a better role function compared to the PG group.
			Souza et al.[27]	Nutritional interventions were able to improve the role function.
	Anxiety/depression	3	Lee S et al.[31]	The depression in BC patients was most severe during NAC compared to before and after NAC. Anxiety was most severe before NAC but gradually improved throughout the treatment session.
			Lee KM et al.[21]	40 patients were anxious and 49 were depressed. In the univariate analyses, overall CINV was significantly associated with anxiety (P=0.036).
			Walker et al.[26]	Anxiety and depression were not statistically significant between the weekly group and the 3-weekly group.

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Subjects	Themes	Study(n)	Author	Outcomes
	Emotional function	4	Chellappan et al.[22]	Three weekly patients had more problems regarding emotional well-being than weekly patients. But the observed difference in scores of depression among the two groups was not statistically significant.
			Rezapour et al.[24]	The PG group had better emotional functioning compared to the AC group.
			Coelho et al.[25]	The emotional function was significantly affected in adjuvant chemotherapy and NAC, but there was no difference in the effects of the two treatments.
			Basen-Engquist et al.[28]	Differences in role emotional were not significant between IG and CG.
Support	Family support	1	Aprilianto et al.[30]	36 respondents (64.3%) were in a good category of family social support. The self-esteem of almost half of the BC patients (27 respondents or 48.2%) was in a moderate level category.
	Social function	4	Chellappan et al.[22]	Weekly group had a high level of social well-being than 3-weekly group.
			Walker et al.[26]	There was no significant difference in the social dimension between weekly group and 3-weekly group.
			Basen-Engquist et al.[28]	Differences in social function of IG and CG groups were not significant.
			Rezapour et al.[24]	The AC group had better social function compared to the PG group.

NAC: neoadjuvant chemotherapy. *QOL*: quality of life. *AC*: doxorubicin and cyclophosphamide. *PG*: paclitaxel and gemcitabine. *CINV*: chemotherapy-induced nausea and vomiting. *IG*: intervention group. *CG:* control group.

Psychological effects

Eight studies evaluated psychological effects during NAC and 52% of patients get negative effects on psychological aspects (Table 2; Figure 2). Four studies evaluated emotional function,[22,24,25,28] three studies evaluated anxiety/depression,[21,26,31] and two evaluated role function.[24,27] Lee S et al.[31] reported that the depression in BC patients was most severe during NAC compared to before and after NAC. Anxiety was most severe before NAC but gradually improved throughout the treatment session. Lee KM et al.[21] reported that overall CINV was significantly associated with anxiety.

Different regimens of neoadjuvant chemotherapy

Three studies investigated the changes in psychological effects with different regimens of NAC. Chellappan et al.[22] studied NAC with different regimens and showed that patients who were treated once a week had a high level of emotional well-being than those who were treated three times a week. But depression among the two groups was not statistically significant. Meanwhile, Walker et al.[26] found anxiety and depression were not statistically significant between the weekly group and the 3-weekly group. Rezapour et al.[24] reported that the AC group had a high level of role functioning than the PG group. The emotional functioning was better in the PG group compared to the AC group.

Neoadjuvant vs adjuvant chemotherapy

One study investigated the psychological effects between NAC and adjuvant chemotherapy. Coelho et al.[25] reported that emotional function was significantly affected in adjuvant chemotherapy and NAC, but there was no difference in the effects of the two treatments.

Intervention in neoadjuvant chemotherapy

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Two studies intervened in BC patients on NAC. Souza et al.[27] reported that nutritional interventions were able to improve the role function. Basen-Engquist et al.[28] reported that weight gain prevention intervention didn't change mental health or role emotion.

Support

Five studies evaluated support from society or family and 55% of patients did not receive adequate support (Table 2; Figure 2). Aprilianto et al.[30] reported that 64.3% of the respondents reported that social support for the patient's family was mostly in the good category, and 48.2% of the BC patients had a moderate level of self-esteem. There was a strong positive correlation between family social support and patient self-esteem.

Different regimens of neoadjuvant chemotherapy

Three studies evaluated support between different regimens of NAC. Chellappan et al.[22] reported that weekly group had a high level of social well-being than 3-weekly group (P=0.007). Rezapour et al.[24] reported that the AC group had significantly better social function compared to the PG group. Walker et al.[26] showed that the weekly group and the 3-weekly group had no significant difference in the social dimension.

Intervention in neoadjuvant chemotherapy

Basen-Engquist et al.[28] reported that weight gain prevention intervention couldn't change social function of patients.

Quality of life-related predictors

Three studies have extrapolated QOL-related predictors of BC patients during NAC: chronotype, QOL before NAC, and depression.[21,29,31] Lee KM et al.[21] suggest late chronotype may be a predictor of CINV. BC patients with late chronotype have an increased risk of CINV during NAC. In another study, Lee S et al.[31] indicated that depression during NAC was a mediator in the relationship between resilience and health-related QOL in BC patients, and QOL after NAC can be improved by screening and intervening for depression during NAC in BC patients. Meanwhile, Takada et al.[29] reported that high QOL before chemotherapy was an independent factor in overall survival. (Table 3)

Table 3 *Quality of life-related predictors*

Study	Sample size(n)	Study design	Instrument	Predictors
Lee KM et al.[21]	134	Prospective cohort study	PSQI, HADS, MAT	Late chronotype may be a predictor of CINV.
Takada et al.[29]	300	Retrospective cohort study	QOL-ACD-B	The level of QOL before chemotherapy was an independent factor in overall survival.
Lee S et al.[31]	193	Prospective cohort study	HADS, FACT-B	Depression was a mediator in the relationship between resilience and health-related QOL.

QOL-ACD-B: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. *FACT-B*: Functional Assessment of Cancer Therapy of Breast. *HADS*: The Hospital Anxiety and Depression Scale. *PSQI*: Pittsburgh Sleep Quality Index. *MAT*: Multinational Association of Supportive Care in Cancer Antiemesis Tool. *CINV*: chemotherapy-induced nausea and vomiting.

Discussion

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In recent years, NAC has been widely used in the treatment of BC, and its efficiency in early-stage BC is as high as 70%~90% and has improved the success rate of breast-conserving surgery and the long-term prognosis of patients in complete pathological remission.[32,33] However, it also has significant side effects on various organs of patients. Patients with BC after NAC require further surgical treatment. Chemotherapy-related symptoms and poor QOL inevitably affect the patient's surgical tolerance and it may delay the patient's surgical treatment in severe cases. The initial stages of treatment and the months following the end of treatment can be difficult times for patients, both physically and emotionally, and BC patients are vulnerable to maladjustment and decreased QOL During these periods.[34] Therefore, information about expected QOL will play an important role in the decision-making process of patients and clinicians considering NAC, so this study was conducted to systematically evaluate the QOL of NAC patients.

The results of this systematic review showed no significant change in overall QOL before and after NAC in BC patients. However, during chemotherapy, patients' QOL decreased, patients experienced increased pain, nausea and vomiting, dyspnea, and fatigue,[20,21,24,25,29] and nausea remained unimproved despite the use of an antiemetic regimen that relieved vomiting.[21] Patients' QOL decreases during chemotherapy due to the toxic effects of chemotherapy drugs, and after chemotherapy, some of the toxic effects diminish and patients' QOL improves. Emotional problems such as anxiety and depression are also aggravated during chemotherapy, but they all resolve at the end of chemotherapy.[25,31] Emotional problems during NAC showed a significant association with resilience before NAC. These findings are largely consistent with previous studies on QOL during NAC for other cancers, such as esophageal cancer,[35] rectal cancer,[36] pancreatic cancer,[37] and muscle-invasive bladder cancer.[38] A transient increase in physical symptoms and a decrease in QOL were seen after neoadjuvant therapy, and in those patients who underwent surgery, most domains returned back by several months.[35–38] Different regimens of NAC have different aspects of impact on patients' QOL. The regimen of doxorubicin, cyclophosphamide, and docetaxel weekly had less distressing side effects compared to the 3-weekly group,[26] and another regimen of paclitaxel and doxorubicin had fewer social well-being compared to weekly group.[22] Comparisons of AC and PG regimens showed that PG had fewer physical adverse effects and AC had better psychological effects.[24] Only one study of included studies compared the impact of NAC to adjuvant chemotherapy on BC patients' QOL,[25] both regimens could affect patients emotionally. However, NAC was more damaging in the terms of physical, cognitive, and personal performance due to more toxicity of the drugs in NAC. QOL can be improved in various ways, including appropriate interventions and enhanced social or family support. Past studies reported that women who are malnourished have a serious negative impact on QOL,[34] and nutritional intervention can improve the symptoms of nausea, vomiting, and loss of appetite in BC patients, and exercise can reduce fatigue and improve the physical quality of BC patients.[27,39,40] Good nutritional status enables patients to better cope with intensive cancer treatment regimens.[41] BC patients with high levels of social support have lower severity of chemotherapy-related symptoms.[42] In addition, interventions can be based on QOL predictors of BC, such as chronotype, the level of QOL before chemotherapy, and depression.

The limitations of this study are mainly in the low inclusion of literature on some sub-topics and insufficient evidence, especially the contrast between NAC and adjuvant chemotherapy. This

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study suffered from publication bias and language bias and was not included some gray literature or unpublished, non-English language literature.

More original studies of high quality are needed. A potential area for further investigations provided by this systematic review is QOL in the perioperative and long-term postoperative periods after NAC. It is important to improve the long-term survival prognosis of NAC but also to focus on and improve QOL during and after NAC.

Conclusion

The results of this systematic review showed that BC patients' QOL was significantly lower during NAC. There was no significant change in overall QOL before and after NAC. And this review found that the influences are mainly reflected in physical symptoms(80%), psychological effects(52%), and support from society or family (55%). QOL could be improved with appropriate interventions. Factors that affect QOL include chronotype, QOL before NAC, and depression can affect the QOL. More original research is needed in the future to understand the profile and predictors of QOL in patients on neoadjuvant therapy, which will help clinicians and patients make better decisions and deal with NAC-related issues.

Footnotes

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Author Contributions: All the authors contributed to the study design. YQ Z and XQ Z screened abstracts and full-text articles for inclusion. YQ Z and LC C appraised study quality. Any disagreements or uncertainties during the screening and quality appraisal process were referred to YL S. YQ Z drafted the manuscript, which was reviewed, edited, and approved by YL S.

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Patient consent: No.

Ethics Approval: This study was a systematic review, so we were not provided the ethics approval.

Data availability statement: All data relevant to the study are included in the article or uploaded as supplementary information.

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Figure 1 Flow chart for systematic review methodology as per PRISMA guidelines

Table 1 Characteristics of included studies

-: not reported. NAC: neoadjuvant chemotherapy. CEF: cyclophosphamide, epirubicin, and 5-fluorouracil. FEC: fluorouracil injection, epirubicin, and cyclophosphamide. AC: doxorubicin and cyclophosphamide. PG: paclitaxel and gemcitabine. NAST: neoadjuvant systemic therapy. IG: intervention group. CG: control group. EORTC QLQ-C30: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30. QLQ-BR23: the Quality of Life Questionnaire-Breast Cancer Module. HADS: The Hospital Anxiety and Depression Scale. PSQI: Pittsburgh Sleep Quality Index. MAT: Multinational Association of Supportive Care in Cancer Antiemesis Tool. QOL-ACD-B: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. HAM-D: Hamilton Rating Scale for Depression. FACT-B: Functional Assessment of Cancer Therapy of Breast. FACT-Taxane: Functional Assessment of Cancer Therapy-Taxane. SF-36: 36-Item Short Form Health Survey questionnaire. BIS: Body Image Scale. RSCL: Rotterdam Symptom Checklist. MRS: Mood Rating Scale. GDS: Global Distress Scale. TSEQ: Treatment Side-Effects Questionnaire. FEDA: Fragebogen erlebter Defizite der Aufmerksamkeit (Questionnaire of Experienced Attention Deficits). RSES: Rosenberg Self Esteem Scale. BFI: Brief Fatigue Inventory. CES-D: Center for Epidemiologic Studies-Depression Scale. HGS: handgrip strength

Figure 2 Summary results of the proportion of each negative impact on quality of life

Table 2 Quality of life of breast cancer patients

NAC: neoadjuvant chemotherapy. QOL: quality of life. AC: doxorubicin and cyclophosphamide. PG: paclitaxel and gemcitabine. CINV: chemotherapy-induced nausea and vomiting. IG: intervention group. CG: control group.

Table 3 Quality of life-related predictors

QOL-ACD-B: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. FACT-B: Functional Assessment of Cancer Therapy of Breast. HADS: The Hospital Anxiety and Depression Scale. PSQI: Pittsburgh Sleep Quality Index. MAT: Multinational Association of Supportive Care in Cancer Antiemesis Tool. CINV: chemotherapy-induced nausea and vomiting.

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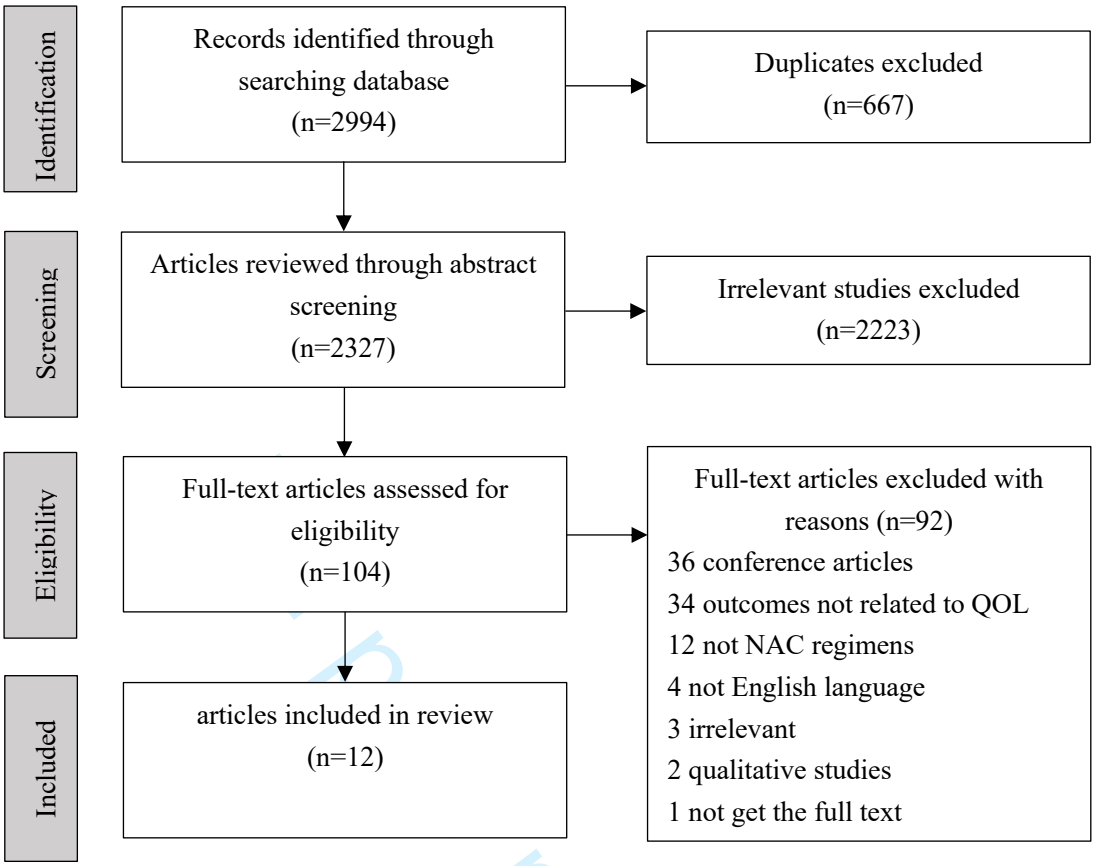


Figure 1 Flow chart for systematic review methodology as per PRISMA guidelines

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Figure 2 Summary results of the proportion of each negative impact on quality of life

peer review only

Supplementary:search strategies

Web of science Search strategy

#4 #1AND #2 AND #3
#3 TS=("Neoadjuvant chemotherapy")
#2 (((((((((TS=("Quality of Life")) OR TS=("life quality")) OR TS=(QOL)) OR TS=(HRQOL)) OR TS=(Support)) OR TS=(depress*)) OR TS=(anxiety)) OR TS=(sexual*)) OR TS=(psychological)) OR TS=(mental)
#1 (((TS=("Breast Neoplasms")) OR TS=("breast tumor")) OR TS=("breast cancer")) OR TS=("breast carcinoma")

Pubmed Search strategy

#8 #4 AND #5 AND #7
#7 #6 OR #2
#6 "psychological"[Title/Abstract] OR "mental"[Title/Abstract] OR "depress*"[Title/Abstract] OR "HRQOL"[Title/Abstract] OR "life quality"[Title/Abstract]
#5 #1OR #3
#4 "Neoadjuvant Therapy"[MeSH Terms]
#3 "breast tumor"[Title/Abstract] OR "breast cancer"[Title/Abstract] OR "breast carcinoma"[Title/Abstract]
#2 "Quality of Life"[MeSH Terms] OR "Social Support"[MeSH Terms] OR "Depression"[MeSH Terms] OR "Anxiety"[MeSH Terms] OR "Sexuality"[MeSH Terms]
#1 "Breast Neoplasms"[MeSH Terms]

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Embase Search strategy

#9 #6 AND #7 AND #8
#8 'neoadjuvant chemotherapy'/exp
#7 'breast tumor'/exp
#6 #1 OR #2 OR #3 OR #4 OR #5
#5 'sexuality'/exp
#4 'depression'/exp
#3 'anxiety'/exp
#2 'quality of life'/exp
#1 'social support'/exp

Cochrane Library Search strategy

#9 #6 AND #7 AND #8
#8 #1 OR #2 OR #3 OR #4 OR #5
#7 MeSH descriptor:[Neoadjuvant Therapy] explode all trees
#6 MeSH descriptor:[Breast neoplasms] explode all trees
#5 MeSH descriptor:[Sexuality] explode all trees
#4 MeSH descriptor:[Depression] explode all trees
#3 MeSH descriptor:[Anxiety] explode all trees
#2 MeSH descriptor:[Social Support] explode all trees
#1 MeSH descriptor:[Quality of life] explode all trees

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Scopus/PsyCINFO/CINAHL Search strategy

#4 #1 AND #2 AND #3

#3 "psychological" OR "mental" OR "anxiety" OR "depress*" OR "quality of life" OR "HRQOL" OR "QOL" OR "life quality" OR "sexual*" OR "support"

#2 "neoadjuvant chemotherapy"

#1 "breast cancer" OR "breast tumor" OR "breast carcinoma" OR "breast neoplasms"

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Supplementary Table S1 *Quality of RCT studies as assessed by Cochrane collaboration's tool for assessing risk of bias*

Study	Random Sequence Generation (selection bias)	Allocation Concealment (selection bias)	Blinding of Participants and Personnel (performance bias)	Blinding of Outcome Assessors (detection bias)	Completeness of Reporting Outcome Data (attrition bias)	Selective Outcome Reporting (reporting bias)	other bias
Walker et al.[26]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Unclear risk
Souza et al.[27]	Low risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk	Low risk
Basen-Engquist et al.[28]	Low risk	Unclear risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk

Supplementary Table S2 *Quality of cohort studies as assessed by Newcastle Ottawa Score(NOS)*

Study	Representativeness of exposed cohort	Selection non-exposed cohort	Ascertainment of exposure	Outcome of interest	Comparability	Assessment of outcome	Length of follow up	Adequacy of follow up	Total score
Ding et al.[20]	*	-	*	-	-	*	*	*	5
Lee KM et al.[21]	*	-	*	-	-	*	*	*	5
Takada et al.[29]	*	*	*	-	**	*	*	*	8
Chellappan[22]	*	*	*	-	**	*	*	*	8
Hermelink et al.[23]	*	-	*	-	-	*	*	*	5
Rezapour et al.[24]	*	*	*	-	*	*	*	*	7
Coelho et al.[25]	*	*	*	-	-	*	*	*	6
Lee S et al.[31]	*	-	*	-	*	*	*	*	6

Supplementary Table S3 *Quality of cross-sectional study as assessed by JBI Critical Appraisal Checklist*

Items	Aprilianto et al.[30]
1. Were the criteria for inclusion in the sample clearly defined?	N
2. Were the study subjects and the setting described in detail?	Y
3. Was the exposure measured in a valid and reliable way?	Y
4. Were objective, standard criteria used for measurement of the condition?	Y
5. Were confounding factors identified?	Y
6. Were strategies to deal with confounding factors stated?	N
7. Were the outcomes measured in a valid and reliable way?	Y
8. Was appropriate statistical analysis used?	Y
Overall appraisal	Include

Y: yes. *N*: no.

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	Page 1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	Page 1
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	Page 2
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	Page 2
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	Page 3
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	Page 2
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	Page 2
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they	Page 3

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		worked independently, and, if applicable, details of automation tools used in the process	
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	Page 3
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	Page 3
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	Page 3
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	Page 3

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Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	Page 3-4
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	Page 3
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	Page 3
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	Page 3
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	n/a.This review could not conduct by meta-analysis.
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	n/a.This review could not conduct by meta-analysis.

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Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	Page 3
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	n/a.This review could not conduct by meta-analysis.
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	Page 3
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	Page 3-4
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	Page 3-4
Study characteristics	#17	Cite each included study and present its	Page 5

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		characteristics	
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	Page 3
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	Page 5
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	Page 5
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	Page 7-11
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	n/a.This review could not conduct by meta-analysis.
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	n/a.This review could not conduct by

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			meta-analysis.
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	n/a.This review could not conduct by meta-analysis.
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	n/a.This review could not conduct by meta-analysis.
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	Page 11-12
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	Page 12
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	Page 12
Implications	#23d	Discuss implications of the results for practice, policy, and future research	Page 12
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	n/a.Registration is still under review.

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Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	n/a.Registration is still under review.
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	n/a.
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	Page 12
Competing interests	#26	Declare any competing interests of review authors	Page 12
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	Page 13

Notes:

- 13e: n/a.This review could not conduct by meta-analysis.
- 13f: n/a.This review could not conduct by meta-analysis.
- 15: n/a.This review could not conduct by meta-analysis.
- 20c: n/a.This review could not conduct by meta-analysis.
- 20d: n/a.This review could not conduct by meta-analysis.
- 21: n/a.This review could not conduct by meta-analysis.

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Quality of life in breast cancer patients with neoadjuvant chemotherapy: a systematic review

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Quality of life in breast cancer patients with neoadjuvant chemotherapy: a systematic review

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Abstract

Objectives The aims of this systematic review were to assess the impact of neoadjuvant chemotherapy (NAC) on breast cancer (BC) patients' quality of life (QOL); to compare the different regimens of NAC on BC patients' QOL; to compare NAC versus adjuvant chemotherapy on BC patients' QOL; and to identify predictors of QOL on BC patients receiving NAC.

Design The design used PRISMA guidelines.

Data sources Cinahl, Embase, Pubmed, Scopus, Web of Science, Cochrane library, and PsycINFO were searched through 27 December 2021.

Eligibility criteria for selecting studies The inclusion criteria were included: BC patients receiving NAC; outcome measures related to QOL; and published in English. The exclusion criteria were included:-duplicates or overlapping participants; not original research; data or full text not available; and-qualitative study.

Data extraction and synthesis Two independent reviewers used standardised methods to search, screen and code included studies. The risk of bias in individual studies was evaluated with Cochrane collaboration's tool for assessing risk bias, Newcastle Ottawa Score, or Joanna Briggs Institute Critical Appraisal tool. This systematic review performs narrative synthesis based on several different themes.

Results The initial search resulted in 2994 studies; 12 of these studies fulfilled inclusion criteria. There was no significant difference in the QOL of BC before and after NAC, but patients experienced adverse reactions and depression during chemotherapy. Different regimens of NAC have different effects on patients' QOL. Patients with NAC had more severe physical discomfort than those with adjuvant chemotherapy. However, BC patients' QOL can be improved by intervening on social or family support, and these predictors including chronotype, QOL before NAC, and depression.

Conclusions More original research is needed in future to understand the profile and predictors of QOL in BC patients on NAC, which will help clinicians and patients make decisions and deal with NAC-related issues.

Other This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Strengths and limitations of this study

►This systematic review was conducted in accordance with PRISMA guidelines.

►Our information sources were 7 research databases from inception through 27 December 2021.

►This review included studies published in the English language only.

Keywords Breast cancer; Neoadjuvant chemotherapy; Quality of life; Systematic review

Introduction

Breast cancer (BC) is the most common disease in women worldwide, with the prevalence now surpassing lung cancer as the leading global cancer incidence in women in 2020, with 11.7% of all new cases (2.3 million); and it is the fifth leading cause of cancer deaths worldwide, accounting for 6.9%.[1] Treatment of BC usually consists of two parts: local surgical treatment and systemic treatment; systemic treatment can precede surgical treatment (neoadjuvant) or follow it (adjuvant).[2] Neoadjuvant chemotherapy (NAC) is used in patients with locally advanced BC or inoperable BC to improve the likelihood of breast-conserving surgery by reducing the size of the tumor in the past. Current studies have shown that NAC can also be used in the treatment of patients with early-stage operable BC to improve cosmetic outcomes and reduce postoperative complications such as lymphedema and prolong survival.[3–5] NAC is associated with improved survival compared with adjuvant chemotherapy in patients with triple-negative BC only after complete pathologic response,[6] and the same effect has been observed in other cancers like bladder cancer and colon cancer.[7,8] NAC is widely used, while more and more scholars are studying the survival outcomes of patients after NAC, such as pathologic complete response, disease-free survival and overall survival.[4,9]

However, the goals of cancer treatment should include not only survival outcomes but also quality of life (QOL), which is an important outcome measure in clinical investigations and survival studies in BC.[10–12] Now the treatment regimens of BC are numerous and effective, but the side effects associated with different treatment modalities and the impact on QOL are different. So optimal stratification and ranking of treatment strategies should focus on individual risk profiles as well as QOL.[13] Meanwhile, QOL is a major factor in treatment decisions for advanced or metastatic disease.[14] It is worthwhile to consider the choice of an effective, least toxic anticancer treatment before implementing definitive therapy. Therefore, studying the QOL of NAC patients and comparing it to other treatments can help patients and clinicians make decisions.

The aims of this systematic review were (1) to assess the impact of NAC on BC patients' QOL; (2) to compare the different regimens of NAC on BC patients' QOL; (3) to compare NAC versus adjuvant chemotherapy on BC patients' QOL; (4) to identify predictors of QOL on BC patients receiving NAC.

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Methods

We primarily followed the Joanna Briggs Institute guidelines[15] on systematic reviews and we used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[16] (Figure 1)

Search strategy

A systematic search of the following databases was performed: CINAHL, Embase, Pubmed, Scopus, Web of Science, Cochrane library, and PsycINFO. The final search was in December 2021. Three categories of terms were searched: (1)breast cancer, (2) neoadjuvant chemotherapy, (3) quality of life. In PubMed and Cochrane library, Medical Subject Headings (MeSH) were used (breast neoplasms, neoadjuvant therapy, quality of life, sexuality, anxiety, depression, social support). In Embase, Emtree terms were exploded (breast tumor, neoadjuvant chemotherapy, quality of life, sexuality, anxiety, depression, social support). In Scopus, Web of Science, CINAHL, and PsycINFO, only keywords were used. (Supplementary material)

Data analysis

The inclusion criteria were as follows: (1) adult patients (18 years older); (2) BC patients receiving NAC; (3) outcome measures related to QOL, including QOL, psychological impact, support from various sources and so on; (4) published in the English language. The exclusion criteria were as follows: (1) data could not be extracted; (2) duplicates or overlapping participants; (3) not original research, such as commentaries, editorials, or reviews;(4) full text not available;(5) qualitative study. Two independent researchers (YQ Z and XQ Z) screened for potentially relevant studies by reviewing abstracts. The full texts were then screened further according to the inclusion and exclusion criteria. Any uncertainty was solved by a third researcher(YL S). Two researchers (YQ Z and XQ Z) independently extracted and recorded the following information from the enrolled studies: author, published year, country, study design, sample size, age, regimens of NAC, timing of assessment, instrument, outcomes. The risk of bias in individual studies was evaluated by 2 independent researchers (YQ Z and LC C) in accordance with Cochrane collaboration's tool for assessing risk bias,[17] or Newcastle Ottawa Score(NOS)[18] or Joanna Briggs Institute Critical Appraisal tool.[19] Any disagreement was resolved by a third researcher(YL S).

Patient and public involvement

Patients or the public were not involved in the conceptualisation or carrying out of this research.

Results

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Flow of included studies

An initial search identified a total of 2994 studies (PubMed: 98, Scopus: 1191, Embase: 281, Web of Science:1011, CINAHL:189, Cochrane library:35, PsycINFO:189). After removing the duplicates, the total number of studies left was 2327 and all of these were screened. On screening, 2223 studies failed to meet our inclusion criteria and were removed from the analysis. The full text was obtained for 104 studies. Out of these 104 studies, an additional 92 were excluded and 12 met our inclusion criteria as defined above. (Figure 1)

Risk of bias in the included studies

The quality of cohort studies was assessed by NOS, RCTs by Cochrane collaboration's tool, and cross-sectional study by Joanna Briggs Institute Critical Appraisal tool. (Supplementary Table S1, S2, S3)

Characteristics of included studies (Table 1)

Seven studies were designed as prospective cohort studies,[20–25] three were randomized controlled trials (RCT),[26–28] one was a retrospective cohort study,[29] and one was a cross-sectional study.[30] Two studies were each from Brazil[25,27] and Korea,[21,31] and one each from China,[20] Japan,[29] India,[22] UK,[26] Germany,[23] Indonesia,[30] Iran,[24] and USA.[28] One study compared QOL in BC patients with NAC and adjuvant therapy,[25] and three studies compared QOL in patients with different regimens of NAC.[22,24,26] Two studies were in the same clinical trial registration with different included subjects.[21,31] The QOL-related instruments used were: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30 (EORTC QLQ-C30), the Quality of Life Questionnaire-Breast Cancer Module (QLQ-BR23), The Hospital Anxiety and Depression Scale (HADS), Pittsburgh Sleep Quality Index (PSQI), Multinational Association of Supportive Care in Cancer Antiemesis Tool (MAT), Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast (QOL-ACD-B), Functional Assessment of Cancer Therapy of Breast (FACT-B), Functional Assessment of Cancer Therapy-Taxane (FACT-Taxane), 36-Item Short Form Health Survey questionnaire (SF-36), Body Image Scale (BIS), Rotterdam Symptom Checklist (RSCL), Mood Rating Scale (MRS), Global Distress Scale (GDS), Treatment Side-Effects Questionnaire (TSEQ), Fragebogen erlebter Defizite der Aufmerksamkeit (FEDA), Rosenberg Self Esteem Scale (RSES), Brief Fatigue Inventory (BFI), Center for Epidemiologic Studies–Depression Scale (CES-D). (Table 1)

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Table 1 *Characteristics of included studies*

Author	Year	Country	Study design	Sample size(n)	Age	Regimens of neoadjuvant chemotherapy	Timing of assessment	QOL-related instrument
Ding et al.[20]	2019	China	Prospective cohort study	29	46±8	CEF regimen	Before and after NAC	EORTC QLQ-C30
Lee KM et al.[21]	2017	Korea	Prospective cohort study	134	44.48±7.48	Anthracycline-based regimens along with standard doses of antiemetics	Before and after the first cycle of NAC	PSQI, HADS, MAT
Takada et al.[29]	2018	Japan	Retrospective cohort study	300	55(27-90)	FEC + paclitaxel(HER2-positive: + trastuzumab)	Before and after NAC	QOL-ACD-B
Chellappan[22]	2018	India	Prospective cohort study	50(weekly) vs 50(3-weekly)	Weekly: ≤50 years accounted for 58%, ≥50 years accounted for 42% 3-weekly: ≤50 years accounted for 56%, ≥50 years accounted for 44%	Fifty patients: paclitaxel and doxorubicin (4 course) every three weekly Other fifty: paclitaxel weekly (10 course) along with doxorubicin (4 course) every three weekly	Weekly (Total 10 weeks)	HAM-D, FACT-B, FACT-Taxane
Walker et al.[26]	2011	UK	RCT	41 (Weekly)vs 41(3-weekly)	Weekly: 50.1(27-68); 3-weekly: 48.3(32-70)	Weekly: AC + weekly docetaxel 3-weekly: AC + 3-weekly docetaxel	Before randomisation, every 3 weeks during docetaxel, and 3 weeks after completion of chemotherapy	FACT-B, RSCL, HADS, MRS, GDS, TSEQ
Hermelink et al.[23]	2007	Germany	Prospective cohort study	101	48.6±9.7	Standard chemotherapy or dose-intensified therapy	Before and toward the end of NAC	FEDA, Cognitive Function Scale of EORTC QLQ-C30; HADS
Aprilianto et al.[30]	2021	Indonesia	Cross-sectional study	56	Almost half (33.9%) of respondents were between the ages of 41 to 50 years old.	-	-	A self-made family social support questionnaire, RSES
Rezapour et al.[24]	2018	Iran	Prospective cohort study	50 vs 50	28~70	AC vs PG regimens	At the beginning and end of chemotherapy	EORTC QLQ-C30
Coelho et al.[25]	2017	Brazil	Prospective cohort study	53(NAC) vs 14(adjuvant chemotherapy)	51.3(30~77)	NAC: anthracyclines and taxanes Adjuvant chemotherapy: -	T1: on the day the treatment was started T2: from 40 to 50 days after the first, when the adverse effects began T3: from 40 to 50 days after the second, when the women were already adapted to treatment or the effects were being handled by the team	EORTC QLQ-C30, QLQ-BR23
Souza et al.[27]	2021	Brazil	RCT	19(IG) vs 15(CG)	44.3±9.2(IG) vs	AC regimen	T0: before the beginning of the first	EORTC QLQ-C30,

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Author	Year	Country	Study design	Sample size(n)	Age	Regimens of neoadjuvant chemotherapy	Timing of assessment	QOL-related instrument
					45.5±8.6(CG)		cycle T1: during the second cycles T2: during the third cycles T3: until the end of the third one	HGS
Basen-Engquist et al.[28]	2020	USA	RCT	19(IG) vs 18(CG)	49.6±13.3 (IG) vs 49.2±9.2 (CG)	-	T0: at baseline T1: mid-chemotherapy (3 months) T2: post-chemotherapy (6 months)	SF-36, BIS
Lee S et al.[31]	2021	Korea	Prospective cohort study	193	44.89±7.913	Four sessions of anthracycline and cyclophosphamide, followed by four sessions of docetaxel.	T0: before the first session of NAC T1: before the start of the last session of chemotherapy T2: 6 months after the end of chemotherapy	HADS, FACT-B

-: not reported. *NAC:* neoadjuvant chemotherapy. *CEF*: cyclophosphamide, epirubicin, and 5-fluorouracil. *FEC*: fluorouracil injection, epirubicin, and cyclophosphamide. *AC*: doxorubicin and cyclophosphamide. *PG*: paclitaxel and gemcitabine. *NAST*: neoadjuvant systemic therapy. *IG*: intervention group. *CG:* control group. *EORTC QLQ-C30*: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30. *QLQ-BR23*: the Quality of Life Questionnaire-Breast Cancer Module. *HADS*: The Hospital Anxiety and Depression Scale. *PSQI*: Pittsburgh Sleep Quality Index. *MAT*: Multinational Association of Supportive Care in Cancer Antiemesis Tool. *QOL-ACD-B*: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. *HAM-D*: Hamilton Rating Scale for Depression. *FACT-B*: Functional Assessment of Cancer Therapy of Breast. *FACT-Taxane*: Functional Assessment of Cancer Therapy-Taxane. *SF-36*: 36-Item Short Form Health Survey questionnaire. *BIS*: Body Image Scale. *RSCL*: Rotterdam Symptom Checklist. *MRS*: Mood Rating Scale. *GDS*: Global Distress Scale. *TSEQ*: Treatment Side-Effects Questionnaire. *FEDA*: Fragebogen erlebter Defizite der Aufmerksamkeit (Questionnaire of Experienced Attention Deficits). *RSES*: Rosenberg Self Esteem Scale. *BFI*: Brief Fatigue Inventory. *CES-D*: Center for Epidemiologic Studies-Depression Scale. *HGS*: handgrip strength.

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Quality of life

Physical symptoms

Ten studies evaluated patients' physical aspects with NAC. NAC affects all aspects of a patient's body and 80% of the patients had adverse physical symptoms (Table 2; Figure 2). Three studies evaluated dyspnea[20,24,25] and pain[26,27,29] respectively, two evaluated nausea/vomit,[21,27] fatigue[20,24] and cognitive function[23,24] respectively, one evaluated body image,[28] and five evaluated other symptoms.[22,24,25,27,29] Lee KM et al.[21] found that 48.5% of patients undergoing NAC experienced chemotherapy-induced nausea and vomiting (CINV), with delayed CINV (prevalence 42.5%) being more common than acute CINV (prevalence 39.6%). Ding et al.[20] reported patients experienced increased dyspnea and fatigue after chemotherapy. Takada et al.[29] found that before NAC, the low QOL group had significantly lower scores on the subscale of Physical symptoms and pain, and the subscale of Dress, sexual aspect, other categories than the high QOL groups (both P<0.001). However, there was no significant difference between the two groups of these subscales after NAC. Hermelink et al.[23] reported during NAC, cognitive function remained stable in most BC patients, but cognitive decline predominated in 27% of patients. It is caused by anxiety and depression from chemotherapy.

Different regimens of neoadjuvant chemotherapy

Three studies investigated the changes in physical symptoms with different treatment regimens of NAC. Chellappan[22] showed that there was no difference in physical problems between the weekly regimen of paclitaxel and the 3-weekly regimen of paclitaxel. Rezapour et al.[24] reported that QOL declined in both doxorubicin and cyclophosphamide (AC) and paclitaxel and gemcitabine (PG) groups, but the two were reflected in different symptoms. The AC group had a significantly better situation in fatigue, pain, cognitive functioning, and constipation compared to the PG group. Nausea and vomiting, dyspnea, insomnia, and diarrhea were better in the PG group. There was no statistically significant difference in physical performance and physical capability between the two groups. Walker et al.[26] reported patients had severe pain in the 3-weekly group compared with the weekly docetaxel group (P=0.034).

Neoadjuvant vs adjuvant chemotherapy

Only one study compared the effects of neoadjuvant and adjuvant chemotherapy on physical symptoms. Coelho et al.[25] observed that QOL was altered with the greater loss for getting NAC compared to adjuvant. Physical function and dyspnea symptoms had a greater impact in the neoadjuvant group compared to the adjuvant group. It is attributed to late diagnoses that provide aggressive treatments.

Intervention in neoadjuvant chemotherapy

Two of these studies intervened in BC patients on NAC. Souza et al.[27] showed that nutritional interventions improved patients' grip strength, positively affected the occurrence of nausea/vomiting and loss of appetite, and reduced the frequency of leukopenia and abdominal pain during NAC. Basen-Engquist et al.[28] reported that weight gain prevention intervention reduced more in waist circumference (P=0.03) and greater improvements in self-reported vitality scores (P=0.03) compared to the control group. The majority of participants reported being satisfied with the

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intervention during chemotherapy (88%).

Table 2 Quality of life of breast cancer patients

Subjects	Themes	Study(n)	Author	Outcomes
Physical symptoms	Body image	1	Basen-Engquist et al.[28]	The difference in body image was not significant between IG and CG.
	Nausea/vomit	2	Lee KM et al.[21]	48.5% of patients undergoing NAC experienced CINV.
			Souza et al.[27]	Nutritional interventions during NAC positively affected the occurrence of nausea/vomiting and loss of appetite.
	Fatigue	2	Ding et al.[20]	Patients experienced increased fatigue after NAC.
			Rezapour et al.[24]	The PG group had increased fatigue than the AC group.
	Cognitive function	2	Hermelink et al.[23]	During NAC, cognitive function remained stable in most patients, but cognitive decline predominated in 27% of patients.
			Rezapour et al.[24]	The PG group had impaired cognitive function than the AC group.
	Dyspnea	3	Ding et al.[20]	Patients experienced increased dyspnea after NAC.
			Coelho et al.[25]	Patients who received NAC had increased dyspnea compared with adjuvant chemotherapy.
			Rezapour et al.[24]	The AC group had increased dyspnea than the PG group.
	Pain	3	Takada et al.[29]	Before NAC, the low QOL group had significantly lower scores on the subscale of Physical symptoms and pain than the high QOL groups(P<0.001). However, there was no significant difference between the two groups after NAC.
			Souza et al.[27]	Nutritional interventions reduced the frequency of abdominal pain during NAC.
			Walker et al.[26]	The pain was significantly greater in the 3-weekly group compared with the weekly group(P=0.034).
	Other symptoms	5	Chellappan et al.[22]	There was no difference in physical problems between the weekly regimen of paclitaxel and the 3-weekly regimen of paclitaxel.
			Takada et al.[29]	Before NAC, the low QOL group had significantly lower scores on the subscale of Dress, sexual aspect, other categories than the high QOL groups(P<0.001). However, there was no significant difference between the two groups after NAC.
			Rezapour et al.[24]	The AC group had a significantly better situation in constipation compared to the PG group. Insomnia and diarrhea were better in the PG group.
			Coelho et al.[25]	The physical function had a greater impact in the neoadjuvant group compared to the adjuvant group.
			Souza et al.[27]	Nutritional interventions improved patients' grip strength, positively affected the loss of appetite, and reduced the frequency of leukopenia during NAC.
Psychological Effects	Role function	2	Rezapour et al.[24]	The AC group had a better role function compared to the PG group.
			Souza et al.[27]	Nutritional interventions were able to improve the role function.
	Anxiety/depression	3	Lee S et al.[31]	The depression in BC patients was most severe during NAC compared to before and after NAC. Anxiety was most severe before NAC but gradually improved throughout the treatment session.
			Lee KM et al.[21]	40 patients were anxious and 49 were depressed. In the univariate analyses, overall CINV was significantly associated with anxiety (P=0.036).
			Walker et al.[26]	Anxiety and depression were not statistically significant between the weekly group and the 3-weekly group.

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Subjects	Themes	Study(n)	Author	Outcomes
	Emotional function	4	Chellappan et al.[22]	Three weekly patients had more problems regarding emotional well-being than weekly patients. But the observed difference in scores of depression among the two groups was not statistically significant.
			Rezapour et al.[24]	The PG group had better emotional functioning compared to the AC group.
			Coelho et al.[25]	The emotional function was significantly affected in adjuvant chemotherapy and NAC, but there was no difference in the effects of the two treatments.
			Basen-Engquist et al.[28]	Differences in role emotional were not significant between IG and CG.
Support	Family support	1	Aprilianto et al.[30]	36 respondents (64.3%) were in a good category of family social support. The self-esteem of almost half of the BC patients (27 respondents or 48.2%) was in a moderate level category.
	Social function	4	Chellappan et al.[22]	Weekly group had a high level of social well-being than 3-weekly group.
			Walker et al.[26]	There was no significant difference in the social dimension between weekly group and 3-weekly group.
			Basen-Engquist et al.[28]	Differences in social function of IG and CG groups were not significant.
			Rezapour et al.[24]	The AC group had better social function compared to the PG group.

NAC: neoadjuvant chemotherapy. *QOL*: quality of life. *AC*: doxorubicin and cyclophosphamide. *PG*: paclitaxel and gemcitabine. *CINV*: chemotherapy-induced nausea and vomiting. *IG*: intervention group. *CG:* control group.

Psychological effects

Eight studies evaluated psychological effects during NAC and 52% of patients get negative effects on psychological aspects (Table 2; Figure 2). Four studies evaluated emotional function,[22,24,25,28] three studies evaluated anxiety/depression,[21,26,31] and two evaluated role function.[24,27] Lee S et al.[31] reported that the depression in BC patients was most severe during NAC compared to before and after NAC. Anxiety was most severe before NAC but gradually improved throughout the treatment session. Lee KM et al.[21] reported that overall CINV was significantly associated with anxiety.

Different regimens of neoadjuvant chemotherapy

Three studies investigated the changes in psychological effects with different regimens of NAC. Chellappan et al.[22] studied NAC with different regimens and showed that patients who were treated once a week had a high level of emotional well-being than those who were treated three times a week. But depression among the two groups was not statistically significant. Meanwhile, Walker et al.[26] found anxiety and depression were not statistically significant between the weekly group and the 3-weekly group. Rezapour et al.[24] reported that the AC group had a high level of role functioning than the PG group. The emotional functioning was better in the PG group compared to the AC group.

Neoadjuvant vs adjuvant chemotherapy

One study investigated the psychological effects between NAC and adjuvant chemotherapy. Coelho et al.[25] reported that emotional function was significantly affected in adjuvant chemotherapy and NAC, but there was no difference in the effects of the two treatments.

Intervention in neoadjuvant chemotherapy

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Two studies intervened in BC patients on NAC. Souza et al.[27] reported that nutritional interventions were able to improve the role function. Basen-Engquist et al.[28] reported that weight gain prevention intervention didn't change mental health or role emotion.

Support

Five studies evaluated support from society or family and 55% of patients did not receive adequate support (Table 2; Figure 2). Aprilianto et al.[30] reported that 64.3% of the respondents reported that social support for the patient's family was mostly in the good category, and 48.2% of the BC patients had a moderate level of self-esteem. There was a strong positive correlation between family social support and patient self-esteem.

Different regimens of neoadjuvant chemotherapy

Three studies evaluated support between different regimens of NAC. Chellappan et al.[22] reported that weekly group had a high level of social well-being than 3-weekly group (P=0.007). Rezapour et al.[24] reported that the AC group had significantly better social function compared to the PG group. Walker et al.[26] showed that the weekly group and the 3-weekly group had no significant difference in the social dimension.

Intervention in neoadjuvant chemotherapy

Basen-Engquist et al.[28] reported that weight gain prevention intervention couldn't change social function of patients.

Quality of life-related predictors

Three studies have extrapolated QOL-related predictors of BC patients during NAC: chronotype, QOL before NAC, and depression.[21,29,31] Lee KM et al.[21] suggest late chronotype may be a predictor of CINV. BC patients with late chronotype have an increased risk of CINV during NAC. In another study, Lee S et al.[31] indicated that depression during NAC was a mediator in the relationship between resilience and health-related QOL in BC patients, and QOL after NAC can be improved by screening and intervening for depression during NAC in BC patients. Meanwhile, Takada et al.[29] reported that high QOL before chemotherapy was an independent factor in overall survival. (Table 3)

Table 3 *Quality of life-related predictors*

Study	Sample size(n)	Study design	Instrument	Predictors
Lee KM et al.[21]	134	Prospective cohort study	PSQI, HADS, MAT	Late chronotype may be a predictor of CINV.
Takada et al.[29]	300	Retrospective cohort study	QOL-ACD-B	The level of QOL before chemotherapy was an independent factor in overall survival.
Lee S et al.[31]	193	Prospective cohort study	HADS, FACT-B	Depression was a mediator in the relationship between resilience and health-related QOL.

QOL-ACD-B: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. *FACT-B*: Functional Assessment of Cancer Therapy of Breast. *HADS*: The Hospital Anxiety and Depression Scale. *PSQI*: Pittsburgh Sleep Quality Index. *MAT*: Multinational Association of Supportive Care in Cancer Antiemesis Tool. *CINV*: chemotherapy-induced nausea and vomiting.

Discussion

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In recent years, NAC has been widely used in the treatment of BC, and its efficiency in early-stage BC is as high as 70%~90% and has improved the success rate of breast-conserving surgery and the long-term prognosis of patients in complete pathological remission.[32,33] However, it also has significant side effects on various organs of patients. Patients with BC after NAC require further surgical treatment. Chemotherapy-related symptoms and poor QOL inevitably affect the patient's surgical tolerance and it may delay the patient's surgical treatment in severe cases. The initial stages of treatment and the months following the end of treatment can be difficult times for patients, both physically and emotionally, and BC patients are vulnerable to maladjustment and decreased QOL During these periods.[34] Therefore, information about expected QOL will play an important role in the decision-making process of patients and clinicians considering NAC, so this study was conducted to systematically evaluate the QOL of NAC patients.

The results of this systematic review showed no significant change in overall QOL before and after NAC in BC patients. However, during chemotherapy, patients' QOL decreased, patients experienced increased pain, nausea and vomiting, dyspnea, and fatigue,[20,21,24,25,29] and nausea remained unimproved despite the use of an antiemetic regimen that relieved vomiting.[21] Patients' QOL decreases during chemotherapy due to the toxic effects of chemotherapy drugs, and after chemotherapy, some of the toxic effects diminish and patients' QOL improves. Emotional problems such as anxiety and depression are also aggravated during chemotherapy, but they all resolve at the end of chemotherapy.[25,31] Emotional problems during NAC showed a significant association with resilience before NAC. These findings are largely consistent with previous studies on QOL during NAC for other cancers, such as esophageal cancer,[35] rectal cancer,[36] pancreatic cancer,[37] and muscle-invasive bladder cancer.[38] A transient increase in physical symptoms and a decrease in QOL were seen after neoadjuvant therapy, and in those patients who underwent surgery, most domains returned back by several months.[35–38] Different regimens of NAC have different aspects of impact on patients' QOL. The regimen of doxorubicin, cyclophosphamide, and docetaxel weekly had less distressing side effects compared to the 3-weekly group,[26] and another regimen of paclitaxel and doxorubicin had fewer social well-being compared to weekly group.[22] Comparisons of AC and PG regimens showed that PG had fewer physical adverse effects and AC had better psychological effects.[24] Only one study of included studies compared the impact of NAC to adjuvant chemotherapy on BC patients' QOL,[25] both regimens could affect patients emotionally. However, NAC was more damaging in the terms of physical, cognitive, and personal performance due to more toxicity of the drugs in NAC. QOL can be improved in various ways, including appropriate interventions and enhanced social or family support. Past studies reported that women who are malnourished have a serious negative impact on QOL,[34] and nutritional intervention can improve the symptoms of nausea, vomiting, and loss of appetite in BC patients, and exercise can reduce fatigue and improve the physical quality of BC patients.[27,39,40] Good nutritional status enables patients to better cope with intensive cancer treatment regimens.[41] BC patients with high levels of social support have lower severity of chemotherapy-related symptoms.[42] In addition, interventions can be based on QOL predictors of BC, such as chronotype, the level of QOL before chemotherapy, and depression.

The limitations of this study are mainly in the low inclusion of literature on some sub-topics and insufficient evidence, especially the contrast between NAC and adjuvant chemotherapy. This

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study suffered from publication bias and language bias and was not included some gray literature or unpublished, non-English language literature.

More original studies of high quality are needed. A potential area for further investigations provided by this systematic review is QOL in the perioperative and long-term postoperative periods after NAC. It is important to improve the long-term survival prognosis of NAC but also to focus on and improve QOL during and after NAC.

Conclusion

The results of this systematic review showed that BC patients' QOL was significantly lower during NAC. There was no significant change in overall QOL before and after NAC. And this review found that the influences are mainly reflected in physical symptoms(80%), psychological effects(52%), and support from society or family (55%). QOL could be improved with appropriate interventions. Factors that affect QOL include chronotype, QOL before NAC, and depression can affect the QOL. More original research is needed in the future to understand the profile and predictors of QOL in patients on neoadjuvant therapy, which will help clinicians and patients make better decisions and deal with NAC-related issues.

Footnotes

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Author Contributions: All the authors contributed to the study design. YQ Z and XQ Z screened abstracts and full-text articles for inclusion. YQ Z and LC C appraised study quality. Any disagreements or uncertainties during the screening and quality appraisal process were referred to YL S. YQ Z drafted the manuscript, which was reviewed, edited, and approved by YL S.

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Data availability statement: All data relevant to the study are included in the article or uploaded as supplementary information.

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Figure 1 Flow chart for systematic review methodology as per PRISMA guidelines

Table 1 Characteristics of included studies

-: not reported. NAC: neoadjuvant chemotherapy. CEF: cyclophosphamide, epirubicin, and 5-fluorouracil. FEC: fluorouracil injection, epirubicin, and cyclophosphamide. AC: doxorubicin and cyclophosphamide. PG: paclitaxel and gemcitabine. NAST: neoadjuvant systemic therapy. IG: intervention group. CG: control group. EORTC QLQ-C30: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30. QLQ-BR23: the Quality of Life Questionnaire-Breast Cancer Module. HADS: The Hospital Anxiety and Depression Scale. PSQI: Pittsburgh Sleep Quality Index. MAT: Multinational Association of Supportive Care in Cancer Antiemesis Tool. QOL-ACD-B: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. HAM-D: Hamilton Rating Scale for Depression. FACT-B: Functional Assessment of Cancer Therapy of Breast. FACT-Taxane: Functional Assessment of Cancer Therapy-Taxane. SF-36: 36-Item Short Form Health Survey questionnaire. BIS: Body Image Scale. RSCL: Rotterdam Symptom Checklist. MRS: Mood Rating Scale. GDS: Global Distress Scale. TSEQ: Treatment Side-Effects Questionnaire. FEDA: Fragebogen erlebter Defizite der Aufmerksamkeit (Questionnaire of Experienced Attention Deficits). RSES: Rosenberg Self Esteem Scale. BFI: Brief Fatigue Inventory. CES-D: Center for Epidemiologic Studies-Depression Scale. HGS: handgrip strength

Figure 2 Summary results of the proportion of each negative impact on quality of life

Table 2 Quality of life of breast cancer patients

NAC: neoadjuvant chemotherapy. QOL: quality of life. AC: doxorubicin and cyclophosphamide. PG: paclitaxel and gemcitabine. CINV: chemotherapy-induced nausea and vomiting. IG: intervention group. CG: control group.

Table 3 Quality of life-related predictors

QOL-ACD-B: Quality of Life Questionnaire for Cancer Patients Treated with Anti-Cancer Drugs-Breast. FACT-B: Functional Assessment of Cancer Therapy of Breast. HADS: The Hospital Anxiety and Depression Scale. PSQI: Pittsburgh Sleep Quality Index. MAT: Multinational Association of Supportive Care in Cancer Antiemesis Tool. CINV: chemotherapy-induced nausea and vomiting.

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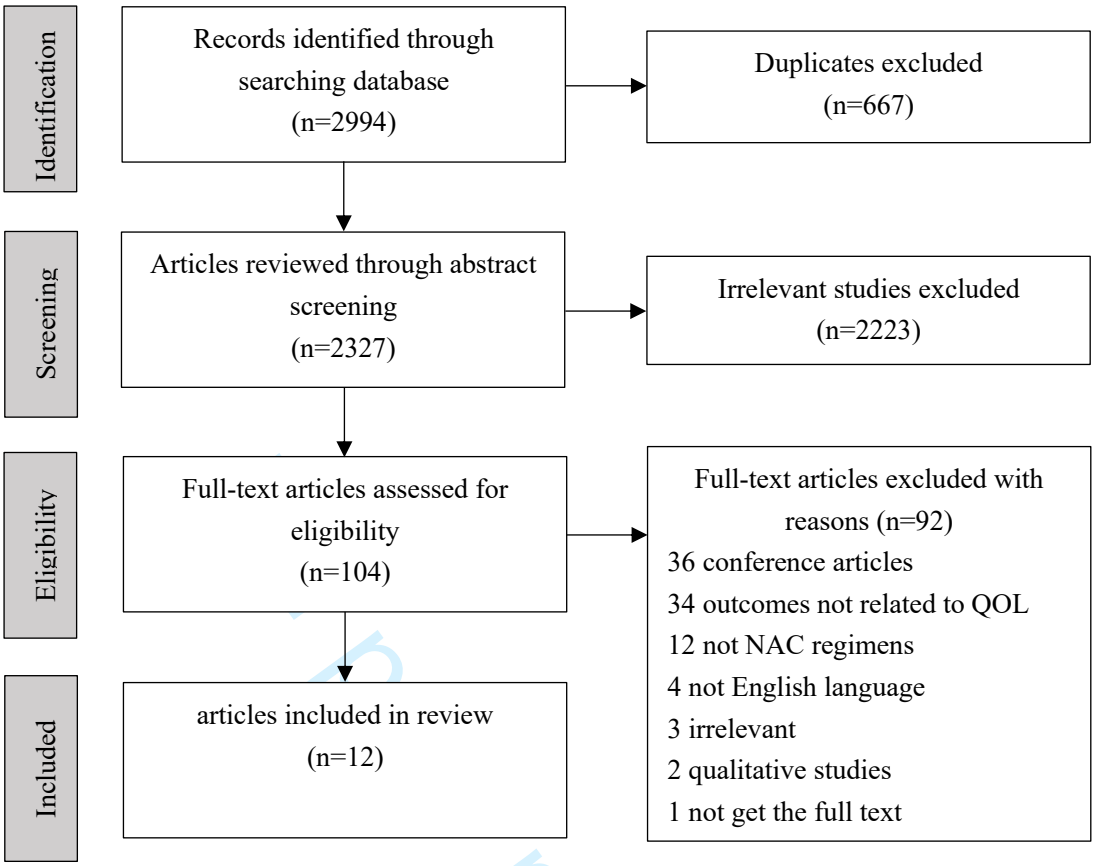


Figure 1 Flow chart for systematic review methodology as per PRISMA guidelines

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Figure 2 Summary results of the proportion of each negative impact on quality of life

peer review only

Supplementary:search strategies

Web of science Search strategy

#4 #1AND #2 AND #3
#3 TS=("Neoadjuvant chemotherapy")
#2 (((((((((TS=("Quality of Life")) OR TS=("life quality")) OR TS=(QOL)) OR TS=(HRQOL)) OR TS=(Support)) OR TS=(depress*)) OR TS=(anxiety)) OR TS=(sexual*)) OR TS=(psychological)) OR TS=(mental)
#1 (((TS=("Breast Neoplasms")) OR TS=("breast tumor")) OR TS=("breast cancer")) OR TS=("breast carcinoma")

Pubmed Search strategy

#8 #4 AND #5 AND #7
#7 #6 OR #2
#6 "psychological"[Title/Abstract] OR "mental"[Title/Abstract] OR "depress*"[Title/Abstract] OR "HRQOL"[Title/Abstract] OR "life quality"[Title/Abstract]
#5 #1OR #3
#4 "Neoadjuvant Therapy"[MeSH Terms]
#3 "breast tumor"[Title/Abstract] OR "breast cancer"[Title/Abstract] OR "breast carcinoma"[Title/Abstract]
#2 "Quality of Life"[MeSH Terms] OR "Social Support"[MeSH Terms] OR "Depression"[MeSH Terms] OR "Anxiety"[MeSH Terms] OR "Sexuality"[MeSH Terms]
#1 "Breast Neoplasms"[MeSH Terms]

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Embase Search strategy

#9 #6 AND #7 AND #8
#8 'neoadjuvant chemotherapy'/exp
#7 'breast tumor'/exp
#6 #1 OR #2 OR #3 OR #4 OR #5
#5 'sexuality'/exp
#4 'depression'/exp
#3 'anxiety'/exp
#2 'quality of life'/exp
#1 'social support'/exp

Cochrane Library Search strategy

#9 #6 AND #7 AND #8
#8 #1 OR #2 OR #3 OR #4 OR #5
#7 MeSH descriptor:[Neoadjuvant Therapy] explode all trees
#6 MeSH descriptor:[Breast neoplasms] explode all trees
#5 MeSH descriptor:[Sexuality] explode all trees
#4 MeSH descriptor:[Depression] explode all trees
#3 MeSH descriptor:[Anxiety] explode all trees
#2 MeSH descriptor:[Social Support] explode all trees
#1 MeSH descriptor:[Quality of life] explode all trees

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Scopus/PsyCINFO/CINAHL Search strategy

#4 #1 AND #2 AND #3

#3 "psychological" OR "mental" OR "anxiety" OR "depress*" OR "quality of life" OR "HRQOL" OR "QOL" OR "life quality" OR "sexual*" OR "support"

#2 "neoadjuvant chemotherapy"

#1 "breast cancer" OR "breast tumor" OR "breast carcinoma" OR "breast neoplasms"

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Supplementary Table S1 *Quality of RCT studies as assessed by Cochrane collaboration's tool for assessing risk of bias*

Study	Random Sequence Generation (selection bias)	Allocation Concealment (selection bias)	Blinding of Participants and Personnel (performance bias)	Blinding of Outcome Assessors (detection bias)	Completeness of Reporting Outcome Data (attrition bias)	Selective Outcome Reporting (reporting bias)	other bias
Walker et al.[26]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Unclear risk
Souza et al.[27]	Low risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk	Low risk
Basen-Engquist et al.[28]	Low risk	Unclear risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk

Supplementary Table S2 *Quality of cohort studies as assessed by Newcastle Ottawa Score(NOS)*

Study	Representativeness of exposed cohort	Selection non-exposed cohort	Ascertainment of exposure	Outcome of interest	Comparability	Assessment of outcome	Length of follow up	Adequacy of follow up	Total score
Ding et al.[20]	*	-	*	-	-	*	*	*	5
Lee KM et al.[21]	*	-	*	-	-	*	*	*	5
Takada et al.[29]	*	*	*	-	**	*	*	*	8
Chellappan[22]	*	*	*	-	**	*	*	*	8
Hermelink et al.[23]	*	-	*	-	-	*	*	*	5
Rezapour et al.[24]	*	*	*	-	*	*	*	*	7
Coelho et al.[25]	*	*	*	-	-	*	*	*	6
Lee S et al.[31]	*	-	*	-	*	*	*	*	6

Supplementary Table S3 *Quality of cross-sectional study as assessed by JBI Critical Appraisal Checklist*

Items	Aprilianto et al.[30]
1. Were the criteria for inclusion in the sample clearly defined?	N
2. Were the study subjects and the setting described in detail?	Y
3. Was the exposure measured in a valid and reliable way?	Y
4. Were objective, standard criteria used for measurement of the condition?	Y
5. Were confounding factors identified?	Y
6. Were strategies to deal with confounding factors stated?	N
7. Were the outcomes measured in a valid and reliable way?	Y
8. Was appropriate statistical analysis used?	Y
Overall appraisal	Include

Y: yes. *N*: no.

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	Page 1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	Page 1
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	Page 2
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	Page 2
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	Page 3
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	Page 2
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	Page 2
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they	Page 3

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		worked independently, and, if applicable, details of automation tools used in the process	
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	Page 3
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	Page 3
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	Page 3
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	Page 3

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Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	Page 3-4
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	Page 3
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	Page 3
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	Page 3
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	n/a.This review could not conduct by meta-analysis.
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	n/a.This review could not conduct by meta-analysis.

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Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	Page 3
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	n/a.This review could not conduct by meta-analysis.
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	Page 3
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	Page 3-4
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	Page 3-4
Study characteristics	#17	Cite each included study and present its	Page 5

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		characteristics	
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	Page 3
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	Page 5
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	Page 5
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	Page 7-11
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	n/a.This review could not conduct by meta-analysis.
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	n/a.This review could not conduct by

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			meta-analysis.
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	n/a.This review could not conduct by meta-analysis.
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	n/a.This review could not conduct by meta-analysis.
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	Page 11-12
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	Page 12
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	Page 12
Implications	#23d	Discuss implications of the results for practice, policy, and future research	Page 12
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	n/a.Registration is still under review.

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Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	n/a.Registration is still under review.
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	n/a.
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	Page 12
Competing interests	#26	Declare any competing interests of review authors	Page 12
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	Page 13

Notes:

- 13e: n/a.This review could not conduct by meta-analysis.
- 13f: n/a.This review could not conduct by meta-analysis.
- 15: n/a.This review could not conduct by meta-analysis.
- 20c: n/a.This review could not conduct by meta-analysis.
- 20d: n/a.This review could not conduct by meta-analysis.
- 21: n/a.This review could not conduct by meta-analysis.

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- 22: n/a.This review could not conduct by meta-analysis.
- 24a: n/a.Registration is still under review.
- 24b: n/a.Registration is still under review. The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 11. February 2022 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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